



MASTER THESIS

Remote Sensing-Based Differentiation of Tree Stress Factors in Urban Environments Using Multispectral and SAR Data: A Case Study in Frankfurt am Main in Co-operation with Karuna Technology

Abstract

Urban trees provide essential cooling, air filtration, and other services but face interacting stressors, drought, nutrient limitations, heat, and soil compaction that threaten vitality. Effective management requires distinguishing stress causes, which most operational remote sensing cannot do.

This thesis develops an integrated remote sensing–machine learning framework to differentiate urban tree stress in Frankfurt am Main, in collaboration with Karuna Technology. High-resolution PlanetScope multispectral data and Sentinel-1 SAR are fused with Arborcheck in-situ physiology (Fv/Fm, chlorophyll content, and SI1–SI4), temporally aligned for selected trees in Ostpark.

A standardized pipeline (radiometric correction, cloud/shadow masking) generates indices (e.g., NDVI, NDRE, CI_{re}, EVI, PRI) and a unified tree-level dataset. Logistic Regression, Support Vector Machine, and Random Forest models classify vitality and stress (healthy, drought-affected, nutrient-related).

Random Forest performs best (accuracy up to 83%), particularly for mild and severe stress, and all models consistently highlight the diagnostic value of red-edge and near-infrared features. Coupling leaf-level diagnostics with satellite observations enables early, city-scale stress differentiation and provides a scalable, cost-effective tool to support targeted interventions, efficient resource allocation, and improved urban climate resilience.

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Remote Sensing-Based Differentiation of Tree Stress Factors in Urban Environments Using Multispectral and SAR Data: A Case Study in Frankfurt am Main in Co-operation with Karuna Technology

by Santosh Ganiger

List of Abbreviations

AOI: Area of Interest

GEE: Google Earth Engine

BBox: Bounding Box

IC: Image Collection

ROI: Region of Interest

SR: Surface Reflectance

NIR: Near-Infrared

RE: Red-Edge

SWIR: Short-Wave Infrared

EVI: Enhanced Vegetation Index

SAR: Synthetic Aperture Radar

NDVI: Normalized Difference Vegetation Index

NDRE: Normalized Difference Red-Edge Index

GNDVI: Green Normalized Difference Vegetation Index

CIGreen: Chlorophyll Index (green band)

MTCI: MERIS Terrestrial Chlorophyll Index

NDWI: Normalized Difference Water Index

VARI: Visible Atmospherically Resistant Index

PRI: Photochemical Reflectance Index

SAVI: Soil-Adjusted Vegetation Index

MSAVI2: Modified Soil-Adjusted Vegetation Index (v2)

RDVI: Renormalized Difference Vegetation Index

SR: Simple Ratio (vegetation vigor metric)

Fv/Fm: Chlorophyll fluorescence efficiency (max PSII efficiency)

VV/VH: SAR polarizations (vertical-transmit/vertical-receive; vertical transmit/horizontal-receive)

MSI: Moisture Stress Index

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1 Introduction

1.1 Context

Urban Heat Islands (UHI's) represent a significant public health risk in dense urban environments. By trapping and re-emitting heat, cities often experience significantly higher temperatures than their rural surroundings, worsening thermal stress. This phenomenon is directly linked to higher rates of human mortality and over mortality. A prominent study by the Barcelona Institute for Global Health (IS Global) on 93 European cities estimated that 4.3% of all summer mortality was attributable to UHI's [39].

A primary strategy for mitigating these hazardous conditions relies on urban trees, which play a vital role as climate-adaptation infrastructure. However, these critical natural assets are themselves highly vulnerable. They face mounting pressures from multiple interacting stressors, including the very intensifying summer heat they are meant to combat, as well as prolonged periods of drought and the constrained, often degraded soils characteristic of urban environments. Recent research has associated extreme heat events and water scarcity with increased crown dieback and higher mortality rates among urban tree populations, highlighting their greater susceptibility compared to trees in surrounding non-urban landscapes [1].

Satellite-based analyses across numerous European cities reveal that urban areas with substantial tree canopy cover maintain significantly lower temperatures than adjacent built-up areas during summer and periods of extreme heat. Typical reductions in land-surface temperature amount to several kelvin, providing clear evidence that tree cover plays a direct role in mitigating urban heat exposure [2].

Modelling studies for European municipalities suggest that increasing average urban tree cover to approximately 30% can lead to measurable reductions in summertime temperatures and prevent heat-related mortality, reinforcing the rationale for canopy expansion as an effective public health intervention [3].

Aside from heat stress, urban settings subject trees to persistent non-climatic pressures that intensify drought impacts. These include compacted, impervious soils that restrict root growth and impede water infiltration, exposure to air pollutants, physical injury, and modified hydrological patterns. Crown condition assessments in both park and streetscape environments reveal that such stressors diminish tree vitality and increase susceptibility to episodic heat–drought episodes.[4] At the physiological scale, deficits in water availability and imbalances in nutrient supply can manifest with similar symptoms, making field diagnosis challenging and potentially delaying the implementation of appropriate interventions such as irrigation or fertilization required for efficient urban tree management. Municipalities often operate with constrained budgets, manage extensive and aging tree populations, and contend with an increasing frequency of heat-related emergencies. These factors create a strong demand for monitoring approaches that are both repeatable and scalable, enabling site triage, the prioritization of interventions, and the evaluation of management outcomes. Remote sensing has become a cornerstone of such workflows, with reviews in urban forestry demonstrating that multisource, multitemporal imagery can deliver consistent, citywide assessments of canopy condition and change at substantially lower costs than manual surveys alone.[5]

Recent case studies focusing on street trees have shown that freely accessible Sentinel-2 time series can capture species-specific drought response patterns within European cities, providing precisely the temporal insights that asset managers require to guide decision-making during periods of water stress.[\[6\]](#)

1.2 Background and Motivation

Urban tree monitoring systems currently implemented by municipalities and service providers typically deliver general assessments of vegetation condition or vitality. While such systems can detect that a tree is under stress, they do not provide information on the specific cause of the stress. This limitation is significant for urban forest management, as the type of stress directly determines the most effective intervention. For example, trees suffering from drought require timely irrigation, whereas those affected by nutrient deficiency require soil amendments or fertilization. Without the ability to differentiate between these stress types, resources may be allocated inefficiently, and the effectiveness of management actions cannot be fully ensured.

This diagnostic limitation primarily stems from the characteristics of the remote sensing indicators most frequently employed. Vegetation indices derived from multispectral data such as the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Water Index (NDWI) are effective in detecting overall variations in canopy greenness and water status [\[7\]](#). Likewise, red-edge indices obtained from Sentinel-2 imagery provide useful insights into chlorophyll levels. However, these indicators are influenced by a range of stressors, including drought, nutrient imbalances, pest damage, and atmospheric pollution. As a result, they cannot, in isolation, pinpoint a single causal factor without additional supporting data [\[9\]](#). Comparative studies of vegetation indices indicate that measures such as the Normalized Difference Red Edge Index (NDRE) and the Chlorophyll Index red-edge exhibit greater sensitivity to variations in chlorophyll concentration and nitrogen content than the NDVI. However, their effectiveness is not uniform and may differ depending on species characteristics and prevailing environmental conditions [\[9\]](#)

The inability to distinguish between different stress types remains a major challenge in urban tree management, especially in the context of increasing climate variability and limited resources. Overcoming this limitation requires approaches that merge the broad spatial and temporal coverage of satellite remote sensing with precise, on-site physiological assessments. By integrating Sentinel-2 multispectral imagery, Sentinel-1 synthetic aperture radar (SAR) data, and ground-based measurements from tools such as Arborcheck which capture chlorophyll fluorescence (F_v/F_m) and chlorophyll content it becomes possible to develop models capable of differentiating between drought-related and nutrient-based stress [\[10\]](#). This capability would enable urban forestry practitioners to deliver targeted treatments and make more efficient use of maintenance resources.

1.2 Research Objectives

The central goal of this research is to design and validate a machine learning approach that can accurately distinguish between drought stress, nutrient deficiency, and healthy states in urban trees by integrating high-resolution satellite data with on-site physiological measurements. This framework aims to overcome the current limitation of many operational monitoring systems,

which can identify general tree stress but cannot determine its specific cause information essential for targeted maintenance decisions.

To fulfil this goal, the study makes use of PlanetScope SuperDove imagery, which provides 8 spectral bands at a spatial resolution of approximately 3 meters suitable for analysing individual trees in dense urban areas. The imagery is processed through a sequence of corrections and filters, including radiometric adjustment, geometric alignment, and the removal of cloud and shadow effects.

Simultaneously, field measurements are collected using the Arborcheck diagnostic system, capturing parameters such as chlorophyll fluorescence efficiency (F_v/F_m), leaf chlorophyll concentration, four stress indices (SI1–SI4), and soil moisture. These field observations are conducted in coordination with satellite acquisition times to maintain temporal consistency between ground and remote data.

The feature development process combines spectral indices derived from PlanetScope data (e.g., NDVI, NDRE, CI_{re}, NDWI) with backscatter metrics from Sentinel-1 radar data (VV, VH, VH/VV ratio), complemented by environmental variables such as rainfall anomalies and soil type where available. All features are harmonized into a unified, tree-specific dataset.

A suite of machine learning algorithms including Logistic Regression, Support Vector Machine, and Random Forest is implemented to classify trees according to their dominant stress condition. Model reliability is evaluated through spatial–temporal cross-validation, and variable importance is analysed to ensure transparency in model interpretation.

The most robust model is then applied to generate detailed, high-resolution stress maps for the study area. These outputs provide spatial insights into tree health, supporting data-driven decisions for municipal green-space management and offering a transferable framework for urban forestry applications in other cities.

2 Literature Review

Monitoring tree health is essential for sustaining urban ecosystems, which provide vital environmental, social, and economic benefits in cities facing mounting climate pressures. Urban forests mitigate heat island effects, sequester carbon, filter air pollutants, and enhance the quality of life for residents. However, trees in dense urban environments are increasingly stressed by a combination of drought, soil nutrient depletion, pollution, and physical damage, often exacerbated by climate change and constrained municipal maintenance budgets. Detecting and diagnosing these stress factors accurately is critical for informed urban forestry management, enabling timely and targeted interventions.

Remote sensing technologies have become an indispensable tool for vegetation monitoring, offering the capability to capture large-scale, repeated measurements of canopy structure, greenness, and water content. Optical sensors such as Planet Scope's 8-band constellation provide high-resolution spectral information across visible, red-edge, and near-infrared wavelengths, enabling the derivation of indices sensitive to chlorophyll content, water status, and general vegetation vigor. Synthetic Aperture Radar (SAR) systems, such as Sentinel-1, complement optical data by penetrating cloud cover and providing structural and moisture-related information through microwave backscatter.

Despite the advances, current operational systems used by municipalities and service providers often only classify trees as "healthy" or "stressed," without identifying the underlying cause. This limits the effectiveness of interventions: drought stress might require irrigation, whereas nutrient deficiency calls for fertilization two very different management responses. Differentiating between these stress types from remote sensing data remains a challenge due to overlapping spectral signatures, seasonal variability, and urban-specific complexities such as mixed pixels, shadows, and heterogeneous species compositions.

Machine learning approaches, including Random Forests, SVM, and Convolutional Neural Networks, have shown promise in extracting subtle spectral-temporal patterns that can distinguish between stress types. When fused with in-situ physiological measurements such as chlorophyll fluorescence efficiency (F_v/F_m), leaf chlorophyll content, and species-specific stress indices remote sensing data can be transformed into powerful predictive models. However, issues such as class imbalance, sensor noise, and the need for robust, interpretable outputs must be addressed to ensure practical utility for city managers.

By integrating multi-source remote sensing with targeted field data, this thesis aims to advance the operational capability to classify urban tree stress types at the individual tree level. This not only supports more efficient use of limited maintenance resources but also contributes to climate adaptation strategies by enabling cities to respond proactively to emerging threats in their green infrastructure.

2.1 Urban tree physiology & common stressors

2.1.1 Water Deficit and Drought Stress

Urban trees operate under a unique combination of abiotic stresses that often exceed those found in rural environments. Water deficit is among the most pervasive, driven by restricted rooting volume, soil compaction, and reduced infiltration from impervious surfaces [11]. Prolonged drought triggers a shift from growth to survival shedding peripheral leaves, reducing fine-root biomass, and lowering photosynthetic capacity. Urban heat island effects amplify water stress through elevated vapor pressure deficits, particularly when maximum temperatures exceed species' thermal tolerance. Severe heat–drought combinations, as documented in Leipzig's 2018–2020 events, can result in widespread canopy thinning and mortality [12].

2.1.2 Nutrient Limitations

Urban soils are often degraded compacted, chemically altered, and low in organic matter leading to nutrient imbalances [13]. Comparative studies on *Acer saccharum*, *Quercus alba*, and *Tsuga canadensis* show higher foliar Ca and Mg in limed urban soils, but consistent deficiencies in K and Mn, which impair stomatal regulation and enzymatic processes [14]. Nitrogen deficiency reduces chlorophyll synthesis and Rubisco activity, resulting in slower growth, chlorosis, and reduced photosynthetic rates.

2.1.3 Visible and Pre-Visual Stress Symptoms

Stress symptoms develop progressively. **Pre-visual indicators** such as chlorophyll fluorescence (Fv/Fm) detect declines in photosystem II efficiency before visible symptoms appear [15]. Additional physiological markers, like leaf chlorophyll content and stress indices (S1–S4), can help differentiate drought-induced photoinhibition from nutrient-related pigment loss. **Visual symptoms** marginal scorch, interveinal chlorosis, necrosis, premature leaf drop emerge once physiological damage surpasses species-specific tolerance thresholds [16].

2.1.4 Morphological and Physiological Acclimation

Some urban trees adjust to heat and drought by altering physiological thresholds (e.g., raising thermal optima for photosynthesis) or morphological traits (e.g., smaller, thicker leaves to reduce transpiration) [17]. However, acclimation capacity varies widely among species, depending on genetic makeup, prior exposure, and local microclimate.

2.1.5 Role of Species Diversity in Resilience

Species diversity particularly the inclusion of drought- and heat-tolerant taxa enhances the resilience of urban forests. Mixed species stands can buffer the effects of extreme climatic

events, reduce pest outbreaks, and support ecosystem stability [18]. This ecological principle is critical for long-term urban tree management strategies.

2.1.6 Synthesis and Implications

Water deficit and nutrient limitations impose distinct, yet sometimes overlapping, physiological constraints on urban trees. Early detection through physiological indicators like Fv/Fm and chlorophyll content enables timely intervention before irreversible decline. Understanding the mechanistic pathways of stress not only informs **in-situ diagnostics** but also supports **remote sensing approaches** where spectral and backscatter signatures can serve as proxies for underlying physiological status.

2.2 Remote sensing of vegetation stress

Healthy plant canopies are characterized by the strong absorption of red light ($\approx 650\text{--}680\text{ nm}$) by chlorophyll and significant reflectance of near-infrared (NIR) light ($\approx 800\text{--}900\text{ nm}$) due to internal leaf scattering. This contrast creates a steep spectral slope around 700 nm , known as the "red edge." Environmental stressors induce physiological changes that alter these spectral properties. For example, a reduction in chlorophyll content heightens reflectance in the red and green portions of the spectrum and shifts the red-edge position toward shorter wavelengths. Furthermore, dehydration typically lowers NIR reflectance by diminishing multiple scattering within the canopy. These spectral responses, especially the high sensitivity in the green-yellow region and near 700 nm , serve as reliable diagnostic indicators of stress across many plant species. [19]

2.2.1 Why multispectral for vegetation stress?

Vegetation "stress" occurs when plants deviate from optimal physiological function due to factors such as drought, nutrient deficiency, disease, heat, or salinity. Optical remote sensing detects these stress signals through variations in pigment composition (chlorophyll and carotenoids), canopy and leaf structure, and water absorption properties. Normalized band ratios commonly known as vegetation indices (VIs) remain widely used because they minimize the influence of illumination differences and sensor calibration drift, allowing for consistent monitoring at high temporal frequency. PlanetScope imagery, with its near-daily global coverage at $\sim 3\text{--}5\text{ m}$ resolution, offers strong potential for field-scale stress mapping despite its limited spectral range and lack of shortwave infrared (SWIR) bands that are critical for assessing plant water content. The newer Super Dove satellites add a red-edge ($\sim 705\text{ nm}$) and two green bands ($\sim 531\text{ nm}$ and $\sim 565\text{ nm}$), enhancing sensitivity to chlorophyll concentration and photosynthetic activity not possible with the earlier four-band sensors. [21]

2.2.2 PlanetScope spectral characteristics

The SuperDove constellation captures imagery across eight spectral bands: **Coastal Blue (443 nm)**, **Blue (490 nm)**, **Green I (531 nm)**, **Green (565 nm)**, **Yellow (610 nm)**, **Red (665 nm)**, **Red-Edge (705 nm)**, and **Near-Infrared (NIR, 865 nm)**. Six of these bands are spectrally aligned with the Sentinel-2 multispectral instrument, while the inclusion of the Green I and Red-Edge bands offers enhanced sensitivity for indices related to chlorophyll concentration and photosynthetic performance, such as the Photochemical Reflectance Index (PRI). In contrast, earlier generations of PlanetScope satellites were limited to four spectral bands Blue, Green, Red, and NIR. The absence of Short-Wave Infrared (SWIR) coverage in all PlanetScope products constrains direct estimation of water-related indices, including the Normalized Difference Moisture Index (NDMI) and the Moisture Stress Index (MSI). Consequently, such indices must be derived through approximation techniques or by integrating PlanetScope data with SWIR-equipped sensors, such as Sentinel-2. [\[21\]](#)

2.2.3 Greenness & canopy structure: NDVI and EVI

The Normalized Difference Vegetation Index (NDVI) serves as a cornerstone remote sensing metric for quantifying vegetation health and vigor. Calculated using the formula $NDVI = (NIR - Red) / (NIR + Red)$, it captures differences in plant reflectance across near-infrared (NIR) and red wavelengths, which correlate with key attributes like leaf area index, chlorophyll concentration, and overall canopy structure. Despite its widespread adoption, NDVI exhibits saturation in areas of abundant biomass, reducing its ability to discern subtle changes in heavily vegetated landscapes.

To address these shortcomings, the Enhanced Vegetation Index (EVI) introduces refinements by integrating data from the blue spectral band alongside adjustment coefficients. These enhancements minimize distortions arising from soil reflectance variations and aerosol effects in the atmosphere, yielding more reliable outputs in diverse environmental settings. In scenarios where blue-band imagery is absent, the EVI2 variant offers a practical substitute relying solely on NIR and red bands while preserving core improvements over NDVI.

Particularly in regions dominated by thick agricultural fields or forested areas, EVI demonstrates a stronger, near-linear correlation with vegetation density compared to NDVI's plateauing behaviour. Pioneering studies from the Moderate Resolution Imaging Spectroradiometer (MODIS) program thoroughly validated the optical properties and ecological linkages of both indices, linking them to metrics like the Fraction of Photosynthetically Active Radiation (FPAR) absorbed by plants and net primary productivity. These established connections hold strong for finer-scale datasets, such as those from PlanetScope satellites, if precise atmospheric and surface corrections are applied to ensure data accuracy. [\[22\]](#)

2.2.4 PlanetScope-Specific Evidence

Recent studies have demonstrated that vegetation indices (VIs) derived from PlanetScope imagery effectively capture crop vigor, yield variation, and management influences at the field scale. Applications include monitoring fertilizer response in winter wheat, in-season yield estimation for corn, and growth assessment in silage maize. Across these investigations, indices such as NDVI, EVI, and related ratio-based metrics (e.g., the Soil-Adjusted Vegetation Index, SAVI) have shown consistent sensitivity to periods of crop stress associated with key phenological transitions. The predictive strength of these indices tends to peak around reproductive growth stages, underscoring their relevance for precision agriculture and early yield forecasting when derived from PlanetScope's high-temporal-resolution imagery.[\[23\]](#)

2.2.5 Water Status: Capabilities and Limitations of PlanetScope

Spectral indices that are physically sensitive to vegetation water status typically require Shortwave Infrared (SWIR) bands. Leaf liquid water exhibits distinct absorption features within the SWIR range, particularly between approximately 1.24 μm and 1.6 μm . Classic indices such as the Normalized Difference Water Index (NDWI; Gao, 1996) and the Moisture Stress Index (MSI; Hunt & Rock, 1989) leverage the spectral contrast between the Near-Infrared (NIR) and SWIR regions to quantify parameters such as equivalent water thickness and relative water content. Because PlanetScope lacks SWIR bands, these indices cannot be computed directly from its data. It is also important to distinguish the McFeeters NDWI (Green–NIR formulation), which was designed for delineating open water bodies, from the vegetation-oriented NDWI; the former should not be used as a proxy for plant water content.[\[24\]](#)

Despite this spectral limitation, several practical workarounds have been demonstrated:

- **Time-series analysis:** Anomalies in NDVI or EVI time series during drying periods can provide indirect evidence of water stress, capturing the vegetation's physiological response rather than the physical water signal itself. Such approaches are frequently employed in drought-monitoring systems, especially when integrated with meteorological or soil-moisture datasets.
- **Physiological indicators:** Indices based on pigment or photosynthetic activity such as the Photochemical Reflectance Index (PRI) and red-edge chlorophyll metrics can reveal early stress responses before visible canopy wilting or NDVI decline, since water deficit often first influences pigment regulation and photosynthetic efficiency.[\[25\]](#)
- **Multi sensor data fusion:** Co-registration of PlanetScope with SWIR-equipped platforms (e.g., Sentinel-2) enables the derivation of water-related indices such as the Normalized Difference Moisture Index (NDMI), MSI, or the Shortwave Infrared Water Stress Index (SIWSI). Ensuring accurate cross-sensor harmonization and Bidirectional Reflectance Distribution Function (BRDF) normalization is critical for maintaining temporal consistency in such fused datasets.[\[26\]](#)

2.2.6 Nutrient and Chlorophyll Indicators from the Red-Edge Region

Chlorophyll content and nitrogen (N) levels are closely linked, making red-edge-based vegetation indices valuable for monitoring crop nutrition. Indices such as the Chlorophyll Index Red-Edge ($CLRE = NIR / Red-edge - 1$) and the MERIS Terrestrial Chlorophyll Index (MTCI) have been extensively validated for estimating canopy chlorophyll content (CCC) across a range of crop types and canopy structures. Multiple reviews highlight that red-edge vegetation indices (VI's) consistently outperform those relying only on red or green wavelengths when estimating CCC or nitrogen status and detecting early stress. With the ~705 nm red-edge band available on PlanetScope's SuperDove sensors, CLRE and similar indices can be calculated directly

Evidence base: Studies by Gitelson and collaborators established the general applicability of red-edge indices for assessing chlorophyll content in crops such as maize and soybean. Follow-up research using various sensors (e.g., MERIS, Sentinel-2) has confirmed that these indices provide more accurate estimates of CCC and often of nitrogen status as well. Comparative analyses further show that indices combining red and red-edge wavelengths or modified simple ratios centered in the red-edge region help overcome signal saturation and enhance sensitivity at high chlorophyll concentrations, where conventional indices like NDVI and EVI tend to level off.[\[27\]](#)

The SuperDove instrument on PlanetScope includes a single red-edge band centered near 705 nm, unlike Sentinel-2, which features three red-edge bands (705, 740, and 783 nm). While this limits the ability to derive metrics based on red-edge position, it still enables reliable use of CLRE-type indices and hybrid vegetation indices that combine the red and red-edge regions. The radiometric stability of SuperDove's spectral bands compares well with that of Sentinel-2, allowing for the creation of consistent multi-year time series when appropriate calibration and processing steps are applied.[\[28\]](#)

2.2.7 Photosynthetic Efficiency and Early-Warning via PRI

The Photochemical Reflectance Index (PRI) measures variations in the xanthophyll cycle, which are directly related to light-use efficiency (LUE) and short-term adjustments in photosynthetic activity. It is calculated as $(R531 - R570) / (R531 + R570)$. Notably, PlanetScope's SuperDove sensors include Green I (531 nm) and Green (565 nm) bands that align closely with the traditional PRI wavelengths, allowing for a reliable approximation of the index (many studies substitute 560–570 nm for the 570 nm band). PRI is highly sensitive to environmental stress such as heat, water, or nutrient limitations and can exhibit measurable responses within minutes, often appearing before NDVI or EVI detect changes. When averaged over daily or weekly intervals, PRI has proven effective for monitoring ecosystem-scale LUE and drought dynamics across diverse vegetation types and biomes.[\[29\]](#)

2.3 Machine Learning for vegetation health

The accurate assessment of vegetation health is a critical imperative for global food security, sustainable forestry, and environmental management. While traditional methods rely on direct fieldwork, they are often insufficient for monitoring large, remote, or rapidly changing ecosystems. Remote sensing, which captures spectral data from satellite or aerial platforms, offers a vast and continuous stream of information. However, translating this raw spectral data into actionable insights about plant physiology requires sophisticated analytical tools. Machine learning (ML) has emerged as the definitive solution for this task, providing robust methods for classification and analysis.[\[36\]](#)

Recent research demonstrates that logistic regression, random forests (RF), and support vector machines (SVM) are effective tools for predicting vegetation health indicators such as NDVI, VHI, and various stress categories using both multispectral and synthetic aperture radar (SAR) data. Logistic regression is valued for its simplicity and interpretability, making it a common baseline model in forest and vegetation health assessments. Random forests are widely adopted due to their resilience to noise and overfitting, achieving strong performance across a range of vegetation types and environmental conditions. Meanwhile, SVM's excel at handling high-dimensional datasets, which makes them particularly effective in hyperspectral stress detection and other complex remote sensing applications.[\[33\]](#)

2.3.1 Machine Learning Classifiers for Spectral Data Analysis

The core task in vegetation monitoring is classification distinguishing healthy pixels from unhealthy ones based on their spectral signatures. The relationship between these signatures and a plant's health status is rarely simple, often requiring models that can capture complex, non-linear patterns. The choice of algorithm involves a trade-off between interpretability and predictive power.

- **Logistic Regression (LR)** is a basic and easy-to-understand method used for classification. It estimates the probability that an input (such as spectral data) belongs to a certain class (for example, "stressed"). The model uses a sigmoid function to convert any real number into a value between 0 and 1, which represents this probability. A threshold, usually 0.5, is then applied to decide the final class. One of the main strengths of Logistic Regression is its interpretability its coefficients clearly show how each feature (like a spectral band) affects the prediction, either positively or negatively. However, a key drawback is that it assumes a linear relationship between the features and the target, which can be too simple for complex patterns in remote sensing data influenced by biological and atmospheric factors. This limitation often leads researchers to use more advanced, non-linear models for better accuracy. [\[30\]](#)
- **Support Vector Machines (SVM)** represent a significant step up in handling non-linear data. The objective of an SVM is to find an optimal "hyperplane" that creates the widest possible margin between different classes. For vegetation analysis, this could be

the boundary that best separates pixels representing 'healthy' crops from those representing 'drought-stressed' crops. The true power of SVM's in this domain is unlocked by the "kernel trick." This mathematical function projects the complex, often inseparable spectral data into a higher-dimensional space where a linear separation becomes possible. This allows the SVM to model highly non-linear relationships without incurring prohibitive computational costs, making it exceptionally well-suited for classifying nuanced spectral signatures that other algorithms might fail to distinguish.[30]

- **Random Forest (RF)** has become one of the most dominant and widely used algorithms for remote sensing tasks. RF is an ensemble learning method, meaning it combines the outputs of many individual models in this case, decision trees to produce a more robust and accurate final prediction. It constructs a large "forest" of uncorrelated trees by training each one on a random subset of the data and considering only a random subset of features at each split. By averaging the predictions of all these diverse trees, the model significantly reduces the risk of overfitting and is less sensitive to noise. This ensemble approach consistently yields very high classification accuracy, often with minimal need for complex parameter tuning.[32]

2.3.2 Challenges in Model Performances

- The challenge of feature selection arises from modern sensor technology. Hyperspectral sensors can capture data in hundreds of spectral bands, and this high dimensionality can degrade model performance by including redundant information. Selecting a smaller subset of the most informative features is therefore crucial. One of the most significant practical advantages of the Random Forest algorithm is its embedded feature selection capability. During training, RF can intrinsically calculate the importance of each feature by measuring how much it contributes to the model's accuracy. This allows researchers to rank all spectral bands by their predictive power and select an optimized subset, leading to a more efficient and accurate model. This built-in function is a key reason for RF's popularity in the remote sensing community.[32]
- The second critical challenge is class imbalance. In any real-world scenario, samples of diseased or stressed vegetation are often far rarer than samples of healthy vegetation. A model trained on such an imbalanced dataset will become biased, achieving high overall accuracy by simply learning to predict the majority 'healthy' class while failing to identify the rare 'unhealthy' class of interest. To combat this, the Synthetic Minority Over-sampling Technique (SMOTE) is a foundational solution. Rather than simply duplicating existing minority class samples, SMOTE creates new, synthetic samples by interpolating between neighbouring instances of the minority class. This process populates the feature space with more, yet varied, examples of the rare class, creating a

more balanced training set that allows the machine learning model to learn the defining characteristics of the minority class.[31]

2.3.3 Feature Selection Strategies

Feature selection is critical when dealing with high-dimensional sensor data (e.g., hyperspectral bands, many VIs, temporal features). Reducing redundant or irrelevant features improves model generalisation and interpretability. The integrated PC–RF–RFE approach combines Pearson correlation analysis with random forest–based feature importance to identify and retain the most informative vegetation indices. The Recursive Feature Elimination (RFE) process then systematically removes features that contribute little to predictive performance, enhancing the model’s ability to generalize. Studies consistently highlight that red-edge, shortwave infrared (SWIR), and chlorophyll-sensitive indices provide the strongest discriminative power for detecting vegetation stress at early stages.[34]

2.3.4 Addressing Class Imbalance

In vegetation stress classification, class imbalance often arises because certain stress conditions such as nutrient deficiencies occur less frequently than others. This imbalance can bias model training and reduce predictive accuracy. To address this issue, researchers commonly apply strategies such as weighted loss functions, data augmentation, and synthetic oversampling techniques like SMOTE to create a more balanced dataset. Additionally, implementing spatial–temporal block cross-validation, as outlined in the referenced thesis framework, helps reduce spatial autocorrelation and data leakage, ensuring more reliable model evaluation in imbalanced remote sensing datasets.[35]

The successful application of machine learning to vegetation health monitoring requires a thoughtful selection of algorithms and meticulous data preparation. While Logistic Regression provides an interpretable linear baseline, the complexities of spectral data often demand more powerful, non-linear classifiers. Support Vector Machines and Random Forest, offer state-of-the-art accuracy and robust performance. However, their potential is only realized when critical data challenges are addressed. The embedded feature selection capacity of Random Forest provides a solution to high dimensionality, while techniques like SMOTE are indispensable for correcting the class imbalance inherent in real-world environmental data. A workflow that integrates these methods provides a powerful pathway for translating raw remote sensing data into actionable intelligence.

2.4 Emerging Methodologies and Identified Gaps

While classical machine learning (ML) algorithms like Logistic Regression, Random Forest (RF), and Support Vector Machines (SVM) are well-established, recent trends show a significant shift towards the adoption of deep learning (DL) methodologies. Architectures such as Convolutional Neural Networks (CNN’s) for spatial pattern recognition and Recurrent

Neural Networks (RNN's), including Long Short-Term Memory (LSTM) variants for time-series analysis, are gaining prominence. A recent review, for example, synthesizes the growing application of DL frameworks for early plant stress detection using complex hyperspectral and temporal datasets

Despite these advancements, several critical gaps remain. A primary limitation is the frequent reliance on data from controlled environments, such as greenhouses or small experimental plots; consequently, the performance of these models in heterogeneous, real-world settings like urban landscapes or mixed-species canopies is often unverified. The transferability of models across different species, sensors, and environmental contexts also remains a significant challenge. Furthermore, the explainability of many DL models is weak, limiting the ability to link algorithm-selected features to underlying biological stress mechanisms. Finally, the synergistic effects of feature selection, methods for handling class imbalance, and data fusion strategies are still under-explored within the domain of vegetation health monitoring.[\[36\]](#)

2.5 Key Challenges in Urban Vegetation Monitoring: Mixed Pixels, Shadows, Heterogeneity, and Species Effects

2.5.1 Background on Urban Vegetation Monitoring

Monitoring the health of vegetation within urban environments which includes disparate elements such as street trees, public parks, and green roofs poses a distinct set of challenges for remote sensing. The urban ecosystem is defined by extreme spatial heterogeneity, not only in the vegetation itself (e.g., variations in species, age, and structural geometry) but also in the underlying soil conditions.

Furthermore, the landscape is a complex mosaic of mixed land cover, where vegetation is interspersed with buildings, concrete, and other impervious surfaces. This arrangement creates significant data contamination from shadows, cast by both anthropogenic structures and adjacent trees. These factors contribute to a high prevalence of mixed pixels, especially in moderate-resolution imagery, complicating spectral analysis. As reviews on the topic emphasize (e.g., Shojanoori et al., 2016, this inherent complexity renders the remote sensing of urban forest health a far more difficult task than monitoring spatially homogeneous areas, such as agricultural plantations or contiguous natural forest stands. [\[37\]](#)

2.5.2 Issues with mixed pixels due to the highly varied urban environment.

A primary challenge in urban remote sensing is the high prevalence of mixed pixels, particularly in satellite or coarse-resolution airborne data. A single sensor pixel in an urban scene rarely captures a pure target; instead, it often integrates spectral information from multiple land-cover types, such as tree canopy, building materials, paved surfaces, soil, and turfgrass.

This spectral mixing has several detrimental consequences for vegetation health monitoring. First, it dilutes the comparatively weak spectral or thermal signal originating from the vegetation, which can mask the subtle signatures of physiological stress. Second, it complicates and confounds automated classification efforts. This is because the spectral variance within a single class (e.g., "healthy tree") increases dramatically; a healthy tree surrounded by dark asphalt will appear spectrally distinct from an identical healthy tree surrounded by irrigated grass, making it difficult for an algorithm to group them.

Several strategies exist to mitigate this problem. The most direct solution is to use higher spatial resolution data, such as from Unmanned Aerial Vehicles (UAVs) or ground-based sensors, which can resolve individual canopies. Other common approaches include object-based image analysis (OBIA), which segments imagery to isolate whole tree canopies before analysis, and spectral unmixing algorithms, which mathematically estimate the proportional abundance of each land cover within a mixed pixel. However, the adoption of high-resolution data introduces its own practical trade-offs, namely a significant increase in data acquisition costs and computational processing loads.

2.5.3 Compounding Factors in Urban Spectral Analysis

Beyond the primary challenge of mixed pixels, several other environmental factors unique to urban settings complicate the detection of vegetation stress.

- **Shadow Contamination:** The complex vertical geometry of the urban environment including "urban canyons" formed by buildings and the proximity of tall, heterogeneously-structured trees results in extensive shadow cover. This shadowing artificially reduces the measured reflectance from a canopy. This can confound stress detection algorithms, as a shaded, healthy tree may produce a "weaker" spectral signal that mimics the signature of a stressed, fully illuminated tree. Correcting for these shadow effects is non-trivial, as their geometry and intensity are dependent on a dynamic interplay of sensor-target geometry, time of day, season, and the specific 3D structure of the objects casting them.
- **Non-Vegetation Backgrounds and Adjacency Effects:** Urban vegetation is frequently surrounded by impervious surfaces such as concrete, asphalt, and building materials, rather than soil. This proximity introduces two types of adjacency effects. Spectrally, these bright or dark backgrounds contribute to the mixed pixel problem. Physically, they alter the microclimate. For example, the high thermal emittance from sun-heated asphalt can increase the surrounding air and surface temperature, which in turn can elevate the canopy temperature of nearby trees and potentially exacerbate water stress.
- **High Species Diversity:** Unlike agricultural monocultures or homogeneous natural forests where most stress models are developed, the urban forest is characterized by high species diversity. This mix of native and exotic species varies significantly in

baseline pigment composition, canopy architecture, phenology, and physiological stress-response mechanisms. This inter-specific heterogeneity makes "one-size-fits-all" stress models, which assume a universal spectral response, highly unreliable. Accurate assessment may therefore require species-specific calibration, yet urban applications must be generalizable across many species. This issue is widely recognized as a fundamental challenge in remote sensing studies involving species classification.[\[38\]](#)

2.5.4 Implications for Methodological Design

The combination of spectral-spatial challenges unique to urban ecosystems has direct implications for the design of effective monitoring strategies, from data acquisition through to final model implementation.

First, to address the high prevalence of mixed pixels and background contamination, data acquisition must prioritize high spatial resolution imagery, such as from Unmanned Aerial Vehicles (UAVs) or terrestrial sensing platforms. This must be paired with robust processing workflows, like object-based segmentation, to accurately delineate and isolate individual tree canopies from the surrounding non-vegetation matrix.

Second, to mitigate false "stress" signals, processing must account for shadow artifacts. This requires the incorporation of radiometric shadow-correction models or the development of algorithms that are robust to illumination variance. Similarly, data acquisition protocols should be standardized, with sensor acquisitions ideally captured under consistent illumination and temporal conditions to reduce confounding spectral variations from diurnal or seasonal cycles.

Furthermore, machine learning models must be explicitly designed to be robust to high environmental heterogeneity. This may necessitate the development of "species-aware" models that incorporate species-specific features or the application of transfer learning and domain adaptation techniques to improve model generalizability across diverse urban contexts.

Finally, these complexities underscore the critical importance of intensive ground-truthing. To properly calibrate and validate remote sensing and ML models, field data collection must move beyond simple visual assessment and instead incorporate tree-by-tree physiological or proximal-sensing measurements.

In summary, when combined with the persistent needs for multi-sensor data fusion and sophisticated class-imbalance handling, these factors confirm that urban tree-health monitoring remains an exceptionally demanding and active domain of research.

2.6 Devices: The Arborcheck System (two devices) Principles, Calculations, and Evaluation

2.6.1 Device Description and Working Principles

The Arborcheck system (Arborcheck Ltd.) is a commercial, portable diagnostic toolset designed for the in-situ assessment of tree vitality and physiological stress. The system is comprised of two principal handheld sensors that conduct measurements on individual leaves: one instrument quantifies chlorophyll content (as a proxy for leaf greenness, like a SPAD meter), and the other measures chlorophyll fluorescence. According to the manufacturer, this dual-assessment approach is designed to detect stress-induced changes at the photochemical level, which often precede the onset of visible symptoms such as chlorosis or wilting.[\[10\]](#)

The working principle of the fluorescence instrument is based on the energy partitioning that occurs within a leaf. In a healthy plant, most of the absorbed solar radiation (reportedly 80–84%) is utilized for photosynthesis. The remaining excess energy must be safely dissipated to prevent photodamage. This dissipation occurs via non-photochemical quenching (NPQ), primarily as heat (infrared radiation) and, to a lesser degree, as re-emitted far-red light, known as chlorophyll fluorescence

When a plant is subjected to stress (e.g., from drought or disease), its photosynthetic efficiency decreases. As the primary energy pathway (photosynthesis) becomes inhibited, the plant must increase its reliance on dissipation pathways. Consequently, a higher proportion of absorbed energy is re-emitted as chlorophyll fluorescence. The Arborcheck system is designed to capture this change: the fluorescence meter quantifies this re-emitted energy as a direct indicator of photosynthetic stress, while the chlorophyll meter assesses the leaf's pigment status, a key indicator of longer-term health and nutrient condition.[\[40\]](#)

2.6.2 What metrics/parameters the devices measure

The first instrument provides a chlorophyll fluorescence metric. While the precise technical specifications of the sensor are proprietary, the measurement functions as a proxy for photosynthetic efficiency. It assesses the partitioning of absorbed light energy between photochemical quenching (photosynthesis) and non-photochemical quenching (dissipation as heat and fluorescence). An elevated fluorescence signal indicates that a lower proportion of energy is being utilized for photochemistry, forcing the leaf to dissipate more energy. This shift is a direct and rapid indicator of physiological stress.[\[10\]](#)

The second instrument is a chlorophyll content meter. This sensor operates analogously to a SPAD meter or "greenness index," providing a quantitative estimate of the chlorophyll concentration within the leaf tissue. This metric is used to identify pigment loss (chlorosis), which serves as a key indicator of longer-term stress responses, potential nutrient deficiencies, or senescence.[\[10\]](#)

To interpret these two measurements, the system includes a proprietary "Nursery Benchmark Database." This database contains baseline optimal vitality ranges for over 400 different tree species. This allows an operator to compare the field data collected from a specific tree against

an expected, healthy benchmark for that species, a feature noted as particularly useful for monitoring the establishment and health of nursery stock or newly planted trees.[\[41\]](#)

2.6.3 Advantages

- **Early Stress Detection:** A primary advantage of this leaf-level approach is the capacity for early, pre-visual stress detection. Physiological parameters, particularly chlorophyll fluorescence and pigment content, are known to respond to stressors before the manifestation of visible symptoms, such as foliar chlorosis or canopy dieback. This capability facilitates proactive management interventions, for instance, following abiotic stress events like soil compaction from construction or suspected root damage.[\[42\]](#)
- **Non-invasive and Rapid Assessment:** The methodology is both non-invasive and rapid. The portable, handheld nature of the instruments permits swift in-situ assessments of individual trees without requiring destructive tissue sampling, which would necessitate laboratory analysis.
- **Species-Specific Benchmarking:** The system's utility is enhanced by the inclusion of a species-specific reference database. This provides crucial context by allowing field measurements to be compared against a baseline of optimal physiological values for a given species, thereby accounting for natural inter-specific variability.
- **Fine-Scale Granularity:** Finally, the system operates at a fine, leaf-level scale. This granularity is particularly advantageous for applications like urban arboriculture or nursery stock management. In these contexts, the health of individual, high-value trees are paramount, and traditional canopy-level remote sensing may be unfeasible or confounded by the landscape heterogeneity and mixed pixels common to urban environments.

2.6.4 Disadvantages

Despite its advantages for fine-scale, pre-visual assessment, this handheld, leaf-level methodology has several critical limitations, particularly concerning its use for broader-scale monitoring.

- **Scalability:** The methodology is constrained by its limited scalability. The reliance on manual, leaf-by-leaf measurements makes its application to large tree populations, such as an entire municipal forest, logistically demanding and resource-intensive. For such broad-scale assessments, this approach is significantly less cost-efficient than automated, canopy-level remote sensing techniques.

- **Species calibration:** The system's accuracy is highly dependent on species-specific calibration. Although the associated benchmark database is extensive, the physiological baselines and stress responses can vary significantly between species. Applying these benchmarks to uncalibrated or non-listed species may therefore introduce considerable inaccuracy.
- **Urban heterogeneity & mixed pixels:** The measurements are characterized by high context sensitivity. A reading from a single leaf reflects only its localized condition and may not be representative of whole-canopy health, especially if stress is localized (e.g., from root damage on one side of a tree). Furthermore, measurements can be easily confounded by sampling inconsistencies, such as variations in leaf age or micro-site illumination (e.g., sampling a sunlit vs. a shaded leaf).
- **Urban heterogeneity & mixed pixels:** When applied to urban settings, the system faces practical and representational challenges. Physical access to canopies for manual measurements can be difficult or hazardous due to tree height. It must also be noted that this point-based tool does not inherently solve the primary challenges of spatial analysis (e.g., spectral mixing, shadow contamination) that are addressed by the remote sensing and machine learning methods discussed previously.
- **Interpretation complexity:** The methodology suffers from ambiguity in causal diagnosis. A detected "stress" signal, such as elevated chlorophyll fluorescence, is often non-specific. Multiple, distinct stressors including drought, nutrient deficiency, soil compaction, and root damage can converge to produce a similar physiological response. Therefore, while the tool can effectively identify that a tree is in a state of reduced vitality, diagnosing the specific underlying cause still requires significant arboriculture expertise and must often be supplemented with additional data, such as soil analysis or nutrient assays.

In summary, the Arborcheck system functions as a valuable proximal sensing instrument within a comprehensive, multi-scale vegetation health monitoring toolkit. Its primary strength lies in targeted, fine-scale inspections where individual tree health is paramount, such as in nursery stock quality control, urban arboriculture, and pre- or post-construction tree health assessments.

The leaf-level physiological data it provides serves a complementary role to broad-area remote and airborne sensing. It can provide granular validation for machine learning models or offer pre-visual stress detection for high-value individual trees. However, due to its manual, point-based nature, it is not a substitute for these remote sensing methodologies when the objective is large-area spatial monitoring or the assessment of entire forest stands.

2.7 Study Area: Frankfurt am Main

Frankfurt am Main, situated in central Germany along the Main River, is the largest urban centre in the state of Hesse and the heart of the Rhine-Main metropolitan region. Home to roughly 770,000 residents, it is Germany's fifth most populous city and functions as a key financial and transportation nexus in Europe. Major institutions such as the European Central Bank, the Deutsche Bundesbank, and Frankfurt Airport—one of the busiest worldwide—underscore its international significance. Alongside its economic prominence and modern skyline, Frankfurt maintains a strong environmental profile, characterized by extensive parks, riverside promenades, and the 3,800-hectare Frankfurt City Forest, which forms part of a protected green belt (“Grüngürtel”) surrounding much of the city. [Figure 2.1](#) provides a spatial overview of Frankfurt, highlighting Ostpark and the surrounding urban fabric. This deliberate balance between urban density and ecological preservation supports Frankfurt's sustainability, quality of life, and climate resilience initiatives. [\[20\]](#) [\[8\]](#).



Figure 2. 1 Spatial Overview of Frankfurt Highlighting Ostpark and Surrounding Urban Fabric. Source: [Google Maps](#)

2.7.1 Climate

The Frankfurt region, situated in central Germany, is characterized by a temperate-humid climate with notable continental influences. This region has not been immune to recent global warming trends; on the contrary, the city has registered some of its warmest months on record. For instance, June 2020 was reported as one of the warmest Junes recorded globally, highlighting the increasing climatic pressures on the region. [\[43\]](#)

For the specific study area of Frankfurt's East End defined here as the districts east of the central ring along the right bank of the Main River, including Ostend, Seckbach, and Bergers Rücken this climate context has several critical implications:

- **Urban Heat Island Effect:** In urban environments, summers are characterized by elevated temperatures, with compact and densely built districts experiencing further intensification due to the UHI effect. This phenomenon is most pronounced during late afternoons and nighttime hours, exacerbating thermal discomfort and energy demands for cooling.
- **Drought Stress:** Despite moderate annual precipitation levels, the increasing frequency and duration of summer dry spells are heightening drought stress. This adversely affects urban vegetation and soil moisture retention, compromising the resilience of green spaces in the face of climate variability.
- **Role of Green Infrastructure:** To mitigate these challenges, green infrastructure encompassing urban forests, parks, green corridors, and street-tree networks emerges as a critical adaptation strategy. These elements deliver vital ecosystem services, such as shading, evaporation cooling, and enhanced stormwater infiltration. Local empirical assessments demonstrate quantifiable benefits, including temperature reductions and improved runoff management, in areas with increased canopy cover and vegetated surfaces.[\[44\]](#)

2.7.2 Soil

The soil and sub-soil conditions of the Frankfurt region are notably heterogeneous, a characteristic derived from its complex geological history, which includes alluvial terraces, fluvial deposits, and glacial sediments. This composition is of relevance to the city's East End.

- A soil profile study for the Frankfurt metropolitan area reveals that the subsoil is frequently composed of sand and gravel layers, which extend to approximately 10 meters below the surface. These layers are subsequently underlain by over-consolidated clay at greater depths. This geological structure is exemplified by the "Kelsterbacher Terrasse," a prominent terrace in Frankfurt's east. It is primarily formed by a core of sand and gravel, which is locally overlain by deposits of loam, clay, or, in some cases, aeolian sand.[\[45\]](#)
- The prevalence of these shallow, coarse-textured soils directly influence arboricultural potential. They naturally limit rooting depth, reduce water storage capacity, and decrease nutrient availability, thereby increasing the vegetation's vulnerability to drought. Within the built-up urban zones, these baseline limitations are further compounded by anthropogenic factors such as soil sealing, compaction, and the disruption of local hydrology, which severely restrict the effective rooting volume and water supply available to trees.

- Furthermore, on a regional scale, recent 3-D mapping of soil carbon stocks within the forest soils of Hesse the federal state encompassing Frankfurt has highlighted the critical role of these organic-carbon-rich soils as carbon reservoirs. The same study also emphasized their high sensitivity to disturbance and management practices.[\[46\]](#)

2.7.3 Land Use and Management Context

Land Use Patterns

The land use in Frankfurt's East End is a complex mosaic, transitioning from high-density residential and commercial zones near the city core to extensive green belts and forested fringe zones on the periphery. This pattern is defined by corridors of green infrastructure, particularly along the Main River and its adjacent open spaces.

This green infrastructure is a cornerstone of the city's identity. Frankfurt maintains one of Germany's largest inner-city forests, the "Frankfurt Stadtwald" (City Forest), which spans approximately 3,866 hectares within the city limits and serves as a critical "green lung". This forest is integrated into the city's "Grüngürtel" (green belt) concept, a strategic planning framework that links forests, parks, and open spaces. This framework ensures high public access to nature; reportedly, 80% of residents live within 300 meters of a public green space. [\[47\]](#) In the East End, this interface between the built environment and the green belt is particularly significant, with areas like the Berger Rücken and Lohrberg representing this critical urban-forest fringe.[\[48\]](#)

Management Context

The responsibility for managing these extensive green assets rests with municipal departments, such as the "Grünflächenamt" (Green Space Office) or "Forstamt" (Forest Office). These agencies oversee all aspects of planning, maintenance, species selection, and urban forest health. A key part of their mandate is to integrate these green spaces into the city's broader strategies for land use and climate adaptation.

This management role involves navigating numerous contemporary challenges. These include the escalating impacts of climate change, such as heat and drought, as well as anthropogenic stressors like soil compaction and limited rooting volumes in built-up areas. Furthermore, managers must balance biodiversity goals, the provision of ecosystem services (e.g., shade, cooling), and intense public-use pressures for recreation.

In the East End specifically, significant land-use pressures exist, creating a core tension between demands for urban densification (housing, transport, commercial use) and the imperative to preserve green spaces. Because the forest and green belt are embedded within a dense urban matrix, all management operations from planting and thinning to hazard tree

removal and storm damage response must be specially adapted. These operations require heightened consideration for public safety, accessibility, and the recreational co-use of these multi-functional landscapes.

2.8 Urban Forest Composition

2.8.1 Dominant Genera

According to the municipal tree inventory, Frankfurt's urban forest exhibits a pronounced dominance of broadleaf species, particularly those well-adapted to the stresses of urban environments such as heat, drought, and pollution. The ten most abundant genera together account for most of the total tree stock. *Acer* (maples) constitutes the largest share with 22.4 % (36,231 trees), followed by *Tilia* (limes) at 9.4 % (15,161), *Carpinus* (hornbeams) at 8.7 % (14,033), and *Quercus* (oaks) at 8.3 % (13,363). Other prominent groups include *Platanus* (planes, 5.7 %), *Fraxinus* (ashes, 4.5 %), *Prunus* (cherries/plums, 4.1 %), *Aesculus* (horse chestnuts, 4.0 %), *Taxus* (yews, 3.3 %), and *Betula* (birches, 2.3 %). This composition reflects a characteristic Central European urban tree palette, combining species valued for their shade provision, cooling effect, aesthetic qualities, and tolerance to pollution. In general, *Quercus* predominates in parks and forest margins, while *Acer*, *Tilia*, and *Platanus* are frequently used in street alignments.[\[49\]](#)

2.8.2 Age Structure

Planting years are documented for nearly all inventoried trees (161,781 entries), allowing a detailed assessment of age distribution. The median planting year is 1984, corresponding to a median age of approximately 41 years (mean $\approx$ 44 years). The age structure is characterised by a strong representation of mid-aged cohorts: trees aged 21–40 years comprise 31.7 % (51,314), and those aged 41–60 years make up 29.9 % (48,314). Younger classes are less numerous, with 11–20 years representing 11.9 % (19,269) and 0–10 years comprising 5.9 % (9,559). Older groups remain well represented, including 61–80 years (11.9 %, 19,301), 81–100 years (5.1 %, 8,222), and over 100 years (3.6 %, 5,802). Overall, roughly 62 % of the trees fall within the 21–60-year range, around 18 % are younger than 20 years, and nearly 9 % exceed 80 years.[\[49\]](#)

2.9 Partner & Operations: Karuna Technology and City Green-Space Management Workflows

Karuna Technology leverages satellite remote sensing and artificial intelligence to offer an innovative platform for urban green infrastructure management. The company's business model is built on delivering superior value through several key factors. These include broad, city-wide spatial coverage, the provision of objective, metric-based data for reporting, and a marked reduction in operational costs. A critical benefit of their system, exemplified by their

ForSens service, is the early detection of vegetation stress (reportedly up to 16 months before visual signs), allowing for proactive intervention in the management of assets like trees, lawns, and other green spaces.

2.9.1 City Greenspace / Urban Forest Management Workflows

1) Inventory and Assessment

This initial phase involves mapping the urban tree population, including details on stand structure, species, age distribution, and current health status or vitality. Traditionally, this relied on field surveys like tree censuses and condition assessments. Modern practices are now integrating new technologies, specifically remote sensing and Artificial Intelligence (AI) tools (such as Karuna Technology's ForSens system), to enhance data collection efficiency and accuracy.

2) Planning and Prioritization

Management goals are formalized in this stage using the collected inventory data. This involves developing strategic plans that outline necessary interventions, which may include new planting schemes, the scheduled removal or thinning of hazard trees, designation of regeneration areas, and adaptation measures to build resilience (e.g., selecting drought-tolerant species or improving soil structure). This phase also covers planning for recreational use and biodiversity enhancement.

3) Implementation and Maintenance

This stage encompasses the practical execution of the developed management plan. Activities involve direct forestry and green-space operations, such as planting, pruning, path maintenance, soil decompaction, mulching, and, critically during dry periods, irrigation. It also includes pest and disease control measures.

4) Monitoring and Feedback

Following implementation, continuous monitoring is crucial to track the success of interventions and the overall health of the urban forest. This involves both on-site field checks and the use of remote sensing data (like that provided by partners such as Karuna). The objective is to track tree health, the consistent delivery of ecosystem services (e.g., shade, carbon sequestration, water infiltration), and the early detection of stress indicators. This data forms a crucial feedback loop, informing the next planning cycle.

5) Adaptation and Reporting

The final component involves the necessary adaptation of management strategies in response to evolving challenges like climate change and increased urban pressure. This includes refining species selection for greater resilience and modifying planting and soil preparation techniques. Furthermore, reporting on ecosystem services and

stakeholder communication (with municipal departments and residents) is vital. Research, such as the BGI findings in Frankfurt, suggests that the most effective management outcomes are achieved through a combination of both "top-down" (municipal-led) and "bottom-up" (community-engaged) activities.

2.9.2 Integration of Karuna Technology with City Workflows

The integration of this remote-sensing and AI-driven technology provides significant advantages for municipal urban forest management. The system's capacity to detect incipient stressors, such as disease, drought, or compromised root systems, functions as an essential early warning mechanism. This predictive capability allows the city to shift from a reactive to a pre-emptive resource allocation model, deploying maintenance teams before issues become critical.

Furthermore, the technology delivers spatially comprehensive data across the entire urban canopy (e.g., in the East End), which can augment or even replace traditional, labour-intensive manual field surveys. This continuous data feed also empowers the municipal green-space department to conduct longitudinal monitoring on the efficacy of its interventions, such as new plantings, canopy thinning, or soil remediation.

Crucially, this data facilitates the quantification of key ecosystem services, including ambient cooling, shade provision, and carbon sequestration, providing tangible metrics for strategic planning and stakeholder reporting. This capability underpins a fundamental shift toward an evidence-based, proactive, and strategically oriented workflow. At a high level, this data-driven approach aligns directly with the city's climate adaptation demands such as mitigating urban heat island effects and monitoring drought risk and reinforces its commitment to stewarding the urban forest as a core sustainability asset.

2.9.3 Practical Considerations for the East End Study Area

Applying this partnership and operational perspective to the East End study area requires a careful review of how municipal workflows must be modified to suit the area's unique conditions. This context includes its status as a dense urban fringe, existing soil limitations, high levels of public use, and the complex boundary between built environments and green spaces.

A collaboration with Karuna Technology could facilitate a pilot program for targeted monitoring within the East End's Forest patches, focusing on areas like mature oak stands and zones of younger regeneration. The goal of this monitoring would be to evaluate vitality trends, identify specific stress hotspots (such as soil compaction zones, vulnerable edge trees, or drought-prone parcels), and provide data to inform management strategies.

You should also propose how management teams can leverage this data to set clear priorities. Examples might include addressing older trees near public trails that pose a risk during storms,

supporting younger stands on sandy soils that require extra irrigation or mulch, or planting more drought-resistant species along terrace soils.

Finally, it is essential to address data governance and operational workflow challenges. This includes clarifying who is responsible for data collection, determining how this information will be integrated into the city's existing GIS and forest inventory systems, and establishing procedures for acting on data-driven alerts. Additionally, a cost-benefit analysis and an assessment of staff training needs should be considered.

2.9.4 Limitations in Current Karuna Workflow and Gaps Addressed by This Thesis

While significant progress has been made in urban forest monitoring, existing systems present several key limitations. This thesis is designed to directly address the following research gaps:

- **Insufficient Ground-Level Validation:** A frequent shortcoming in remote sensing is that satellite and AI-generated health indicators (like vitality indices) often lack robust verification from on-site physiological data. This research bridges that gap by correlating direct, ground-truth data (such as Arborcheck physiological metrics including Fv/Fm, chlorophyll content, and S1-S4 stress indicators) with remotely sensed information. This linkage serves to verify and calibrate the metrics used to assess vegetation health from space.
- **Lack of Local Specificity:** Broad, generic models frequently fail to account for unique, site-specific variables. Factors such as the heterogeneous soil types, distinct microclimates, and diverse species composition found in the East End may not be adequately captured. This study overcomes this deficiency by developing relationships calibrated specifically for the site that connect local environmental factors, spectral data, and the measured health of the trees.
- **Challenges in Temporal Resolution and Early Detection:** This work also confronts issues related to the speed of detection. While some platforms provide monthly data updates, physiological stress in trees, particularly early in the season, can manifest more rapidly than spectral changes are captured by satellites. By integrating high-frequency, in-situ measurements, this research evaluates the timeliness and sensitivity of satellite-based indicators, aiming to improve the early detection of tree stress.
- **The Operational Integration Gap:** A significant gap often exists between data provision and its practical application. Advanced data products from platforms like Karuna's are available, but their routine adoption and integration by municipal staff remains a developing process. A key component of this thesis is to evaluate how such

data streams can be effectively embedded within established city workflows to support and enhance adaptive management practices.

- **Uncertainty in Multi-Scale Analysis:** Finally, this thesis tackles the inherent uncertainty involved in scaling data. A persistent challenge lies in accurately translating stress measurements taken at the individual tree or leaf level (e.g., Arbor check metrics) to broader canopy- or landscape-scale indices (e.g., pixel-level satellite data). This study actively tests and assesses various scaling methodologies to bridge this divide and improve the consistency between ground-level observations and landscape-wide monitoring.

3 Implementation and Methodology

3.1 Tree Selection and Equipment Setup

The practical implementation of this study began with the collaboration of the Grünflächenamt Frankfurt am Main (Municipal Green Space Department), which played a central role in facilitating field access and supporting the selection of representative urban trees for physiological data collection. Based on the objectives of this research to develop a diagnostic model for differentiating drought and nutrient stress in urban trees the Grünflächenamt identified and designated eight trees for in-situ measurement within Frankfurt's Ostpark, one of the city's principal green areas.

The selection process prioritized species that are both ecologically and municipally significant, representing dominant taxa in Frankfurt's urban forest composition. The chosen species included:

- ***Ilex aquifolium* (Common holly)**
- ***Fagus sylvatica* (European beech)**
- ***Carpinus betulus* (European hornbeam)**
- ***Quercus robur* (Pedunculate oak)**
- ***Tilia cordata* (Small-leaved lime)**

These species collectively encompass a range of physiological and morphological traits, from evergreen to deciduous habits and from shade-tolerant to light-demanding ecotypes. This diversity was intended to ensure that the collected field data reflected varying adaptive strategies and stress responses, providing a robust empirical basis for model development.

Following tree selection, the Grünflächenamt provided the research team with a complete Arborcheck system, consisting of two main portable instruments:

- A Chlorophyll Content Meter, which quantifies the relative chlorophyll concentration within leaf tissue, and
- A Chlorophyll Fluorescence Meter, designed to assess photosystem II efficiency as an indicator of photosynthetic performance and physiological stress.



Figure 3. 1 Arborcheck field kit

The equipment was supplied in a calibrated and fully operational state, as shown in [Figure 3.1](#) (Arborcheck field kit). The case included both handheld meters, a tablet preloaded with the Arborcheck data interface, USB charging adapters, and accessory cables for device connectivity. The Grünflächenamt provided an initial demonstration and technical guidance on how to properly handle and operate the devices in field conditions, ensuring standardized measurement procedures for all subsequent data collection campaigns.

The Ostpark site was selected due to its accessibility, species diversity, and relevance within the city's broader green infrastructure. It represents a semi-natural, yet highly managed environment, typical of the urban-forest-park interface found across Frankfurt's green belt. The trees sampled were spatially distributed across different microhabitats of the park to capture potential variability in soil moisture, exposure, and canopy cover factors that can significantly influence stress responses.

The first phase of implementation thus established the experimental foundation of the study: defining the sample population, ensuring equipment readiness, and aligning fieldwork protocols with municipal standards. Subsequent stages focused on the systematic collection of physiological and environmental data using these instruments, synchronized with satellite acquisition periods, as detailed in the following sections.

3.2 Field Data Collection Procedure

Field data collection was carried out in Frankfurt Ostpark using the Arborcheck system, consisting of two main handheld instruments: a Chlorophyll Fluorescence Meter and a Chlorophyll Content Meter, both connected to the Arborcheck tablet interface. The purpose of this phase was to obtain in-situ physiological measurements that could serve as ground-truth data for satellite-based analysis. Measurements were conducted exclusively during morning

hours, between 09:00 and 12:00, aligning with the approximate overpass time of PlanetScope satellites, to ensure temporal consistency between ground and remote sensing datasets.

3.2.1 Preparation and Clip Placement

For each selected tree, ten sampling points were chosen around the canopy to represent the full physiological variability of the individual. Leaves were selected from different orientations and heights (where accessible) to capture both sun-exposed and shaded conditions, thus avoiding bias toward a single canopy microenvironment.

Special Arborcheck leaf clips ([Figure 3.2](#)) were used to prepare the leaves for fluorescence measurement. Each clip consists of a circular cover that isolates the leaf surface from ambient light, allowing for accurate measurement of dark-adapted photosystem activity. The clips were attached gently to the leaves as shown in Figures [3.2](#) and [3.3](#), ensuring that the leaf tissue was not damaged and that all sampled leaves were healthy and mature.

A total of ten clips were evenly distributed around each tree, following a circular pattern to encompass all canopy orientations. After attachment, each clip was kept in place for a minimum of 30 minutes to allow for dark adaptation, a critical step that ensures the photosynthetic apparatus reaches a steady dark-acclimated state suitable for chlorophyll fluorescence analysis.



Figure 3.2 Arborcheck clip clipped to leaf for measurement.



Figure 3. 3 Clipping the Arborcheck clip to leaves for measurement

3.2.2 Measurement of Chlorophyll Fluorescence

After the 30-minute adaptation period, measurements were conducted using the Arborcheck Chlorophyll Fluorescence Meter ([Figure 3.4](#)). The device was positioned directly against each clip, and the protective cap was briefly opened to expose the leaf to a pulse of measuring light. This instrument records the maximum quantum efficiency of photosystem II (F_v/F_m), a physiological indicator of photosynthetic performance and stress.

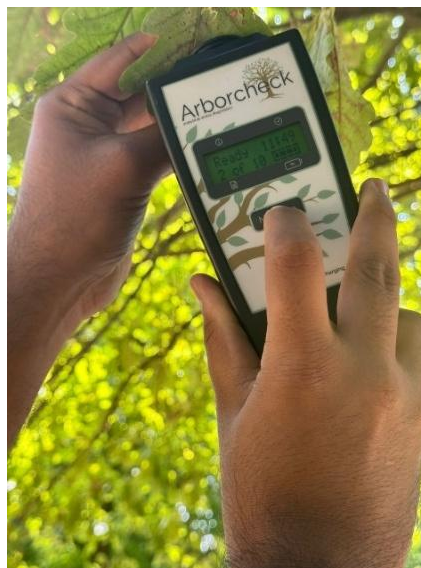


Figure 3. 4 Measuring chlorophyll fluorescence using the Arborcheck device.

In healthy, unstressed leaves, F_v/F_m values typically range between 0.78 and 0.84, whereas deviations below this range indicate potential physiological stress caused by drought, nutrient imbalance, or other environmental factors. For each tree, ten such fluorescence readings were collected one for each clipped leaf and automatically averaged by the device.

All readings were stored internally and later synchronized via Bluetooth with the Arborcheck application on the tablet for digital processing and data management.

3.2.3 Measurement of Chlorophyll Content

Following fluorescence measurements, the same ten leaves were analysed using the Chlorophyll Content Meter ([Figure 3.5](#)). The clips were carefully removed, and the sensor head was positioned directly against the same leaf area previously used for fluorescence measurement. This instrument quantifies leaf chlorophyll concentration, which serves as a proxy for nutrient status and long-term physiological health. Each reading was displayed instantly on the meter and manually recorded into the Arborcheck application. As with fluorescence data, chlorophyll readings were collected for all ten leaves per tree and averaged to obtain a representative value for that individual tree.



Figure 3. 5 Measuring chlorophyll content using the Arborcheck device.

3.2.4 Data Recording and Synchronization

Once the measurements were completed for each tree, all fluorescence data were automatically transferred via Bluetooth to the Arborcheck application installed on the field tablet. The tablet interface ([Figure 3.6](#)) allowed for the input of metadata including Tree ID (Baum ID), species name, and GPS coordinates, ensuring spatial traceability and proper linkage between ground observations and remote sensing pixels.

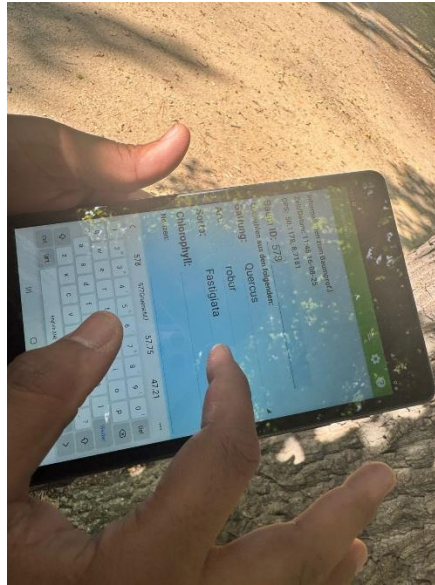


Figure 3. 6 Logging field data through the Arborcheck tablet application.

The chlorophyll content values obtained from the second instrument were entered manually (Figure 3.7) into the tablet interface to merge both physiological datasets under the same tree record. After all data fields were completed, the record was submitted within the application, triggering the automated computation of the Arborcheck vitality profile (Figures 3.8 & 3.9).



Figure 3. 7 Entering chlorophyll measurement data in the Arborcheck app.

3.2.5 Data Output and Interpretation

The Arborcheck system immediately generated a vitality summary for each tree. The interface displayed a colour-coded vitality indicator ranging from green (optimal health) to red (high stress), along with four stress indices (SI1–SI4), which quantify deviations from species-specific benchmark values stored in the Arborcheck reference database. These indices

correspond to different physiological pathways such as photochemical efficiency, pigment stability, and non-photochemical quenching responses.

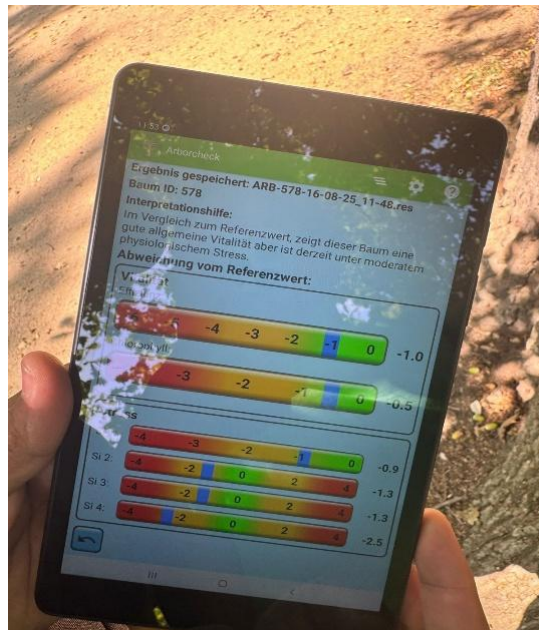


Figure 3. 8 Displaying chlorophyll fluorescence, chlorophyll content and stress results in the Arborcheck application.



Figure 3. 9 Displaying overall tree vitality assessment in the Arborcheck application.

Each recorded dataset was saved in a “.res” (result) file format, which contains both the raw measurement values and the derived indices. These files were later exported from the tablet for data cleaning, aggregation, and integration into the subsequent analysis pipeline, where they served as ground-truth inputs for remote sensing model calibration.

3.3 Tree Setup and Satellite Data Preparation

After the completion of ground-based measurements, the next step in the implementation phase involved preparing spatial references for each sampled tree to facilitate the extraction of satellite-derived spectral data. The primary objective of this process was to link each individual tree's Arborcheck physiological measurements with its corresponding PlanetScope spectral reflectance data, ensuring a precise spatial match between field and satellite datasets.

3.3.1 Defining Tree Locations in Google Earth Engine (GEE)

The Google Earth Engine (GEE) platform was utilized to manually geolocate and delineate each sampled tree. GEE offers a cloud-based environment for geospatial data visualization, processing, and integration, which made it particularly suitable for aligning field observations with high-resolution satellite imagery.

Using the satellite basemap and available PlanetScope imagery layers within GEE, each tree location was identified based on its known GPS coordinates collected during the field survey. Once identified, a polygon was drawn around the tree canopy to define its Area of Interest (AOI). These polygons serve as spatial boundaries for subsequent data extraction of spectral band values.

To create polygons in GEE, the following general workflow was followed:

- Access the GEE code editor at <https://code.earthengine.google.com>
- Open the interactive map panel and navigate to the approximate coordinates of each tree using the search bar or by manually zooming to the Ostpark study area.
- Select the “Geometry tools” icon in the map toolbar.
- Choose the polygon drawing tool and outline a small area that accurately represents the canopy extent of the tree, ensuring it corresponds with the location where Arborcheck data were collected.
- Assign an attribute label (e.g., BaumID or Tree ID) to each polygon to maintain consistency with field data records.
- Repeat the process for all sampled trees until all polygons are defined and visible in the map layer. ([Figure 3.10](#))

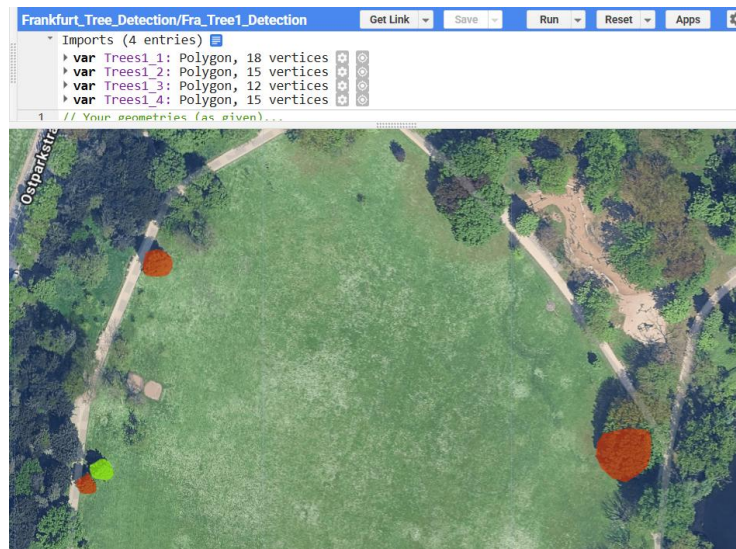


Figure 3. 10 Tree polygon delineation in Google Earth Engine (GEE) for the Ostpark study area

The final set of polygons was saved as geometries within the GEE script environment, where each was assigned a unique feature identifier (FID) and its corresponding BaumID, ensuring a one-to-one relationship between field and satellite datasets.

3.3.2 Feature Collection and Shapefile Export

After defining all tree geometries, each polygon was converted into a feature containing both its geometry and associated attributes (FID and BaumID). These features were compiled into a single FeatureCollection, representing the complete spatial dataset for the sampled trees.

To enhance usability in subsequent stages, additional attributes such as polygon area and centroid coordinates (latitude and longitude) were automatically computed for each feature. These derived attributes support future integration with QGIS or Python-based geospatial analyses.

The completed FeatureCollection was then exported as a shapefile (.shp) to Google Drive, ensuring compatibility with other GIS environments used later for data extraction and preprocessing. The exported shapefile contained only essential fields FID and BaumID to maintain simplicity and alignment with field data identifiers.

The full GEE implementation script, including all procedural steps for attribute creation and shapefile export, is available at the following link:

<https://code.earthengine.google.com/5c305ce62bdec2f9d301a7bdeb513b45>

3.4 Satellite Image Preparation and Bounding Box Generation

The acquisition of PlanetScope satellite imagery required a precise spatial setup to ensure that the downloaded data corresponded accurately with the tree locations identified during fieldwork. All images were downloaded via the Planet API, with access credentials provided

by Karuna Technology. Since the actual image retrieval script is proprietary to Karuna, this section provides a descriptive overview of how the process was implemented and how the bounding box (BBox) and related data were generated in Google Earth Engine (GEE).

3.4.1 Creating an Image Collection in GEE

Before initiating the download process, an Image Collection (IC) was created within the GEE Assets.

In GEE, an Image Collection serves as a container that holds multiple satellite scenes or processed images under a single reference. Creating a collection involves the following steps:

- Open Google Earth Engine at <https://code.earthengine.google.com>
- In the Assets panel, click on “NEW” → “Image Collection”.
- Assign a unique collection name (e.g., GDE_8B_FRA_PSScene2) and specify the storage folder or directory within your GEE account.
- This empty collection then acts as the destination for PlanetScope images imported later through the API-based retrieval process.

By storing the Planet imagery in an Image Collection, it becomes easier to manage, filter, and process multiple temporal scenes in subsequent analysis steps.

3.4.2 Planet API Inputs

Downloading PlanetScope imagery through the Planet API requires three key parameters:

- 1) **API Key** – This is a unique authentication token provided by Karuna Technology through Planet Labs. It allows authorized access to Planet’s data repository and API services. According to Planet Labs’ documentation, this key must accompany all API requests to verify the user’s credentials and permissions to access imagery.
- 2) **Bounding Box (BBox)** – The BBox defines the rectangular geographic extent (spatial boundary) of the area of interest for which imagery is to be downloaded. It ensures that all selected trees within the study area are covered by the retrieved images.
- 3) **Corner Points** – These are the four geographic vertices (north-west, north-east, south-east, and south-west) of the bounding box. They are critical for defining the exact limits of the download area and ensuring accurate spatial alignment between the Planet imagery and the ground coordinates collected during field measurements.

The use of a precise BBox and its corner points is crucial when dealing with high-resolution PlanetScope data (3–5 m spatial resolution). Even minor deviations in spatial boundaries could result in misalignment between the downloaded imagery and the target trees.

3.4.3 Bounding Box Generation Process

The bounding box generation was performed within Google Earth Engine (GEE). The region of interest (ROI) corresponding to the tree locations in Frankfurt Ostpark was used as the starting geometry.

The process began by simplifying the ROI polygon to generate a clean and minimal BBox that encompassed the entire study area. The script calculated and printed the area of the ROI to confirm the extent, and then generated a simplified BBox geometry, which was visually added to the GEE map interface for verification.

Once the bounding box was confirmed, its four corner points were automatically computed. Each corner point was represented as a small polygon to facilitate later integration with the Planet API. This ensured that the download area was accurately aligned with the selected trees and their surrounding canopy area.

The intermediate results BBox area, simplified polygon, and corner coordinates were printed in the GEE console to verify spatial consistency. Once validated, a function was executed to combine all the relevant details (BBox geometry, corner coordinates, and metadata) into a single JSON configuration string.

3.4.4 Output Configuration and Interpretation

The final output of this process was a structured JSON-like configuration file, which contained all spatial and temporal metadata required for image retrieval.

The configuration file defines:

- The image collection name where the data will be stored in GEE,
- The temporal schedule for when downloads or updates could be executed,
- The bounding box geometry, and
- The four corner polygons defining its spatial limits.

Each element ensures a structured and automated connection between the Planet API and GEE, reducing manual effort and potential for spatial mismatches.

3.4.5 Verification and Preparation for Image Download

Once the bounding box and corner polygons were generated, they were verified visually in the GEE map view to ensure full coverage of the Ostpark study area and the sampled trees. The output JSON was then used in the subsequent Planet API script (executed via Karuna's proprietary toolset) to automate image retrieval.

3.4.6 Image-Collection (IC) Download Script Logic and Workflow

This subsection explains how the IC download script ingests PlanetScope scenes into the pre-created GEE ImageCollection (e.g., GDE_8B_FRA_PSScene2). The actual script is proprietary; the description below documents its function and processing logic without exposing code.

a) Authentication and Configuration

- API authentication: The script authenticates to the Planet Labs API using the API key supplied by Karuna Technology.
- Region & storage: It loads the AOI package prepared earlier (Section [3.4.3–3.4.5](#)): project short name, the minBBox polygon, and its four corner polygons. It also points to the target GEE Asset ImageCollection where scenes will be stored.
- Product & bundle: The script requests PlanetScope SuperDove (8-band) PSScene surface reflectance as the primary product, together with the UDM2 quality mask (clouds, haze, shadow, saturation).
- Timing: For each field campaign date, it defines a day-window around the visit to prioritize temporal proximity to in-situ measurements.

b) Scene Search and Filtering

- Spatial filter: The minBBox is used as the search footprint so only scenes intersecting the Ostpark study area are considered.
- Temporal filter: Searches are constrained to the specified start–end dates for each campaign window.
- Quality screen: Scenes are filtered using Planet metadata (e.g., overall cloud fraction, acquisition quality flags). Only scenes meeting pre-set thresholds are retained.
- De-duplication: If multiple scenes cover the same date/time, preference is given to the one with the best quality metrics (lower cloud/haze, best off-nadir angle).

c) Order Creation, Activation, and Download

- Order batching: Eligible scenes are grouped into orders (by date window and AOI) to respect Planet API limits while minimizing overhead.
- Asset activation: For each scene, the script requests the SR + UDM2 assets and polls until they are “active.”
- Retrieval: Once active, assets are downloaded to a temporary workspace with their original metadata (acquisition time, spacecraft ID, sun angles, etc.).

d) Pre-ingestion Harmonization

- Projection & georeferencing: Scenes are verified to use the expected CRS and 3-m pixel size; any deviations are corrected so that all scenes align on a common grid.

- Band layout: Bands are standardized to a consistent order and naming (coastal blue, blue, green I, green, yellow, red, red-edge, NIR).
- Radiometry: Surface-reflectance scale factors from Planet metadata are applied (where required) so all scenes are in physically comparable units.
- Quality mask: The UDM2 mask is parsed into per-pixel Boolean layers (cloud, cloud-shadow, haze, saturation). A combined “valid data” mask is created for later use.

e) Ingestion to Google Earth Engine

- Upload to IC: Each scene is ingested as a single Image into the target GEE ImageCollection.
- Properties: The script writes key properties onto every Image (e.g., acquisition_datetime, scene_id, cloud_percent, sun_zenith, off_nadir, provider=Planet, product=PSScene_SR, aoi_id=FRA_2).
- Mask attachment: The corresponding UDM2-derived mask is included so that downstream processing in GEE can transparently exclude invalid pixels.
- Naming convention: Asset IDs follow a consistent pattern (collection name + acquisition timestamp), enabling simple temporal filtering later.

f) Post-ingestion QA/QC

- Coverage check: The script cross-checks that each ingested Image intersects the minBBox and reports spatial coverage over the tree polygons.
- Cloud audit: A summary table (date, scene id, cloud %, valid pixel %) is logged for each acquisition window to document why a scene was kept or skipped.
- Visual verification: For spot checks, the script adds the latest ingested scene and its mask to the GEE map, overlaid with the Ostpark polygons, to verify alignment.

g) Output

- A populated GEE Image Collection containing 8-band PlanetScope SR scenes for the Ostpark AOI, each accompanied by a UDM2-based validity mask and standardized metadata.
- A lightweight ingestion log (dates, scene IDs, QA metrics) used later to select scenes for feature extraction and modelling.

This completes the image-acquisition pipeline: authentication → search & filter → order/activate/download → harmonize → ingest to GEE IC → QA/QC. Subsequent sections describe how this curated collection is clipped to the tree polygons and transformed into analysis-ready features.

3.4.7 Post-Processing of the Downloaded Image Collection

After the PlanetScope image scenes were successfully imported into the Google Earth Engine (GEE) Image Collection, a dedicated post-processing workflow was applied to prepare the data for analytical use.

This stage ensured that every image was radiometrically consistent, spatially aligned, and filtered to include only high-quality, cloud-free pixels covering the Ostpark study area.

a) Data Harmonization and Metadata Verification

Each PlanetScope scene in the Image Collection was first checked to confirm that it contained the expected eight surface-reflectance bands (coastal blue, blue, green I, green, yellow, red, red-edge, near-infrared). Scene-level metadata such as acquisition date, spacecraft ID, solar and view geometry, and cloud-cover percentage were parsed from the Planet metadata fields and attached as properties to each image.

This step ensured that subsequent temporal and spectral operations could rely on standardized metadata across all acquisitions.

b) Radiometric and Atmospheric Consistency

Although PlanetScope Level-3B products are distributed as surface-reflectance data, minor differences in scale factors and bit depth can exist between individual scenes. The post-processing script automatically normalized all reflectance values to a uniform physical range (0 – 1) by applying the Planet-provided scale factors.

In addition, any residual haze or atmospheric scattering effects detected from the UDM2 quality mask were corrected using a simple haze-removal adjustment that subtracts background path radiance from each spectral band.

c) Cloud and Shadow Masking

Quality assurance (QA) masks supplied through the UDM2 layer were used to exclude pixels contaminated by clouds, cloud shadows, or saturation. The script converted these QA bits into a Boolean “valid data” mask and applied it to all bands in each image.

This produced a clean set of reflectance composites free from atmospheric or illumination artefacts. Where necessary, small data gaps caused by masking were interpolated using the temporal median of neighbouring dates within a ± 3 -day window to preserve spatial continuity.

d) Spatial Alignment and Clipping to Area of Interest

Each cleaned image was re-projected to the common coordinate reference system used in the project (EPSG: 4326) and resampled to 3 m pixel resolution. The scenes were then clipped to the previously defined minimum bounding box (minBBox) surrounding the Ostpark trees. This clipping step reduced file size and ensured that every pixel within the collection corresponded exactly to the spatial extent of the study area.

e) Temporal Aggregation and Compositing

To reduce short-term variability while retaining seasonal dynamics, the processed images were aggregated into temporal composites.

For each observation window, the script computed:

- Median composites, representing typical reflectance conditions and minimizing outliers from residual noise.
- Maximum NDVI composites, capturing peak vegetation activity where appropriate; and
- Mean reflectance composites, used for spectral-index computation.

This yielded a temporally balanced dataset aligned with the timing of the field campaigns.

f) Spectral-Index Generation

From the harmonized reflectance bands, several vegetation and stress-related indices were calculated directly in GEE, including:

- Normalized Difference Vegetation Index (NDVI) – proxy for canopy greenness and photosynthetic activity.
- Normalized Difference Red-Edge (NDRE) – sensitive to chlorophyll concentration and early stress.
- Green Normalized Difference Index (GNDVI) – linked to leaf nitrogen content; and
- Simple Ratio (SR) – alternative measure of vegetation vigour.

Each index was appended as a new band to the corresponding image within the Image Collection, allowing multivariate analysis in later stages.

g) Export and Data Archiving

Once all indices were computed and quality-checked, the processed collection was exported and archived in GEE under a same Imagecollection. Metadata logs were automatically generated, summarizing acquisition date, valid-pixel percentage, and index statistics (mean, median, standard deviation) for each scene. This ensured complete traceability and reproducibility of the dataset.

The resulting post-processed PlanetScope Image Collection provided a fully calibrated, cloud-free, and analysis-ready dataset for subsequent integration with the field-measured physiological parameters collected using the Arborcheck system.

3.4.8 Spectral Band Data Extraction for Individual Trees

After preparing the post-processed PlanetScope image collection ([Section 3.4.7](#)), the next stage focused on extracting spectral band information for each individual tree polygon. This

operation is critical because it translates pixel-based remote sensing data into tree-specific spectral metrics that can later be correlated with the field-measured physiological parameters obtained using the Arborcheck system.

The extraction workflow was implemented in Google Earth Engine (GEE) using a custom script developed by Karuna Technology. The proprietary code automates the retrieval, filtering, averaging, and export of spectral reflectance values for each tree geometry.

a) Input Datasets

Two key datasets were required to perform the extraction:

1) Post-Processed PlanetScope Image Collection

This dataset represents the final, cleaned, and harmonized imagery obtained from Planet Labs after atmospheric correction and cloud masking. It contains eight spectral bands (B1–B8) corresponding to the PlanetScope SuperDove configuration:

- B1: Coastal Blue (430–450 nm)
- B2: Blue (465–515 nm)
- B3: Green I (513–549 nm)
- B4: Green (547–583 nm)
- B5: Yellow (600–620 nm)
- B6: Red (650–680 nm)
- B7: Red Edge (697–713 nm)
- B8: Near Infrared (845–885 nm)

The dataset is stored as a GEE Image Collection (e.g., GDE_8B_FRA_PSScene2) and represents temporally stacked surface reflectance scenes covering the study area.

2) Tree Polygon Shapefile

This dataset contains the vector geometries for all sampled trees in the Ostpark site. Each polygon delineates the canopy area of a single tree, as previously digitized in GEE ([Section 3.3](#)).

The attribute table includes:

- FID: unique feature ID within the shapefile.
- BaumID: identification number corresponding to field data.
- Species name (Gattung) and Tree number (Baumnummer) descriptive attributes derived from the field database.

These polygons form the spatial basis for calculating average spectral reflectance values at the individual tree level.

b) Initialization and Visualization

Once both datasets were loaded into the GEE workspace, the tree polygons were displayed as vector overlays and the map view was centered on the Ostpark area. This visual verification step ensured that the polygon geometries aligned correctly with the underlying satellite scenes. This was crucial before running the extraction since even small spatial misalignments could cause erroneous reflectance sampling from surrounding pixels rather than the actual canopy.

c) Defining Spectral Bands and Filtering the Image Collection

A list of band identifiers (B1–B8) was defined to target the eight PlanetScope reflectance channels for extraction. The script then applied a sequence of filtering operations to select only the most relevant and high-quality images from the Image Collection:

- Spatial filtering: Only images intersecting each tree polygon's geometry were retained. This limits the analysis to scenes that contain the target area.
- Temporal filtering: The date range was restricted to the month of August 2025, corresponding to the field campaign period. This temporal alignment ensures that satellite observations represent similar environmental and phenological conditions to those captured in the field.
- Quality filtering: Several metadata-based filters were applied to ensure that only cloud-free, high-confidence imagery was used:
 - I. `Local_clear_percent` ≥ 0.99 – ensures at least 99% of the scene was clear.
 - II. `local_mean_clear_confidence_percent` ≥ 80 – keeps only high-confidence clear pixels.
 - III. `publishing_stage` = finalized – excludes provisional or unverified scenes.
 - IV. `quality_category` = standard – ensures only operational-quality imagery is used.
 - V. `image_order_in_date` = 1 – ensures that only the first, most relevant acquisition per date is selected.

This rigorous filtering eliminated noisy or partially cloudy images, ensuring the spectral data extracted for each tree were spatially and temporally consistent.

d) Per-Tree Processing and Band Reduction

The core operation of the script involved looping through each tree polygon in the shapefile and calculating mean reflectance values for all eight bands.

The process for each tree was as follows:

- 1) Identify intersecting images: For each tree polygon, the script identified all PlanetScope images overlapping the polygon area within the filtered Image Collection.
- 2) Reduce spectral data: For each selected image, the mean pixel value of each band (B1–B8) was calculated over the polygon area using a zonal statistics reducer (mean reducer). The pixel scale was fixed at 3 meters PlanetScope's native resolution

ensuring that canopy-level details were preserved without oversampling or aggregation errors.

- 3) Create a feature record: A new feature (record) was generated containing:
 - The FID and BaumID of the tree,
 - The date of the satellite image,
 - The mean reflectance values (B1–B8), and
 - Tree details, concatenating the species and tree number attributes for easy identification.
- 4) Store feature results: Each resulting feature was appended to a temporary FeatureCollection, which accumulates the spectral summaries for all trees.

Through this approach, the workflow produced a detailed multi-date spectral database where each entry represents one tree observed on one date

e) Batch Processing for Large Datasets

Since the dataset contained numerous tree polygons, the extraction was performed using a batch-processing system to optimize computational performance. The shapefile was divided into groups of 1,000 features per batch, and the script processed each batch sequentially. This approach ensured stable execution within GEE's resource limits and allowed continuous monitoring of processing progress. After processing each batch, its output was merged into the main results collection, ensuring data integrity across all trees.

f) Data Export and File Structure

After processing was complete, each batch of extracted results was automatically exported to Google Drive as a Comma-Separated Values (CSV) file.

The exported CSV files contained the following columns:

- 1) **FID**: Unique feature identifier for the tree polygon
- 2) **BaumID**: Field tree identification number
- 3) **Date**: Acquisition date of the PlanetScope image
- 4) **TreeDetails**: Combined text string of species and tree number
- 5) **ShapeID**: Internal GEE identifier for the polygon
- 6) **B1–B8**: Mean reflectance values of each PlanetScope band

Each exported batch followed a consistent naming convention (e.g., FraOst_Planet_Tree_Data.csv), making it easy to merge them into a single unified dataset later.

Once exported, the CSV files were downloaded and consolidated locally for further analysis in Python and R.

g) Purpose and Outcome of the Extraction

The final output of this process was a tree-level spectral reflectance dataset, providing the average reflectance across all eight PlanetScope bands for each sampled tree on each available image date. This dataset forms the essential bridge between field-based physiological measurements (chlorophyll content, fluorescence) and remotely sensed reflectance data.

By correlating these reflectance values with ground-based Arborcheck readings, it becomes possible to develop predictive models and spectral indicators that estimate physiological stress in urban trees directly from satellite imagery.

h) Advantages of the Automated Workflow

This automated GEE-based extraction offers several key advantages:

- **Spatial precision:** Ensures reflectance is computed only from the tree canopy area, avoiding background contamination.
- **Temporal synchronization:** Guarantees that imagery dates correspond closely to field data acquisition.
- **Scalability:** The batch-based approach enables efficient processing of large urban tree inventories.
- **Standardization:** Automates quality control and ensures consistent data extraction across multiple trees and time periods.

Through this step, the remote sensing component of the study was effectively linked to ground observations, enabling further analysis of urban tree health based on multispectral satellite data.

3.5 Reading, Cleaning, and Transformation of Arborcheck Data

This stage of the workflow focused on preparing and standardizing the field-based physiological data collected through the Arborcheck system. These data serve as the ground truth for calibrating and validating the satellite-based analysis of tree health.

The Arborcheck system generates outputs in a proprietary format (.res) that combines raw sensor readings with contextual metadata. Therefore, the first step involved systematically reading, cleaning, and transforming these files into a consistent, analysis-ready format.

The resulting dataset was later integrated with the PlanetScope satellite-derived reflectance data to create a unified input for machine learning model training.

3.5.1 Structure and Interpretation of Arborcheck .res Files

The Arborcheck instrument stores every measurement session in an individual .res file. Each file contains two parts: a numerical data section consisting of raw sensor channels and a metadata section providing contextual information such as tree ID, species, coordinates, and device references.

A sample .res file used in this study is shown below:

```
5
-0.7338380789730774,1
0.0,1
-0.6859398449556923,1
-1.5871696091094656,1
-1.4111645590240554,1
-0.23954517132221384,1
749,17,8,25,9,43,Quercus,robur,*MEASURED*,"","","","","","",2010 to 2018
(A),07-2019-01-A-200004.abd,50.1213287000,8.7223002400
13004
```

The content of the file can be explained as follows:

- 1) Header Line (5): Indicates the number of raw measurement records contained in the file. In this example, five lines of raw data follow.
- 2) Raw Channel Values

Each of the next five lines contains two values separated by a comma:

- The first value represents the sensor reading, typically a normalized or calibrated fluorescence intensity or reflectance signal from the leaf surface.
- The second value is a binary quality flag: “1” means valid, “0” means invalid. These values represent the raw signal response from the leaf before any physiological interpretation is applied by the Arborcheck software.

For instance:

```
-0.7338380789730774, 1
0.0, 1
-0.6859398449556923, 1
```

These indicate multiple sequential sensor samples taken from the same leaf, all flagged as valid (flag = 1).

- 3) Metadata Line: This is the most information-rich part of the file, containing descriptive attributes separated by commas:

```
749,17,8,25,9,43, Quercus, Robur,*MEASURED*,"","","","","","",2010 to 2018
```

The fields are as follows:

Position	Example Value	Description
1	749	BaumID: Unique tree identifier assigned by Grünflächenamt
2	17	Day of measurement
3	8	Month
4	25	Year (two-digit form → 2025)
5	9	Hour (local time, 24-hour clock)
6	43	Minute
7	Quercus	Genus
8	robur	Species epithet
9	MEASURED	Measurement status indicating a successful reading

This line therefore links the measurement to a specific tree, species, location, and calibration configuration used by Arborcheck.

- 4) Footer Line (13004). Represents an internal record identifier generated by Arborcheck to uniquely label each measurement session. This value ensures traceability across multiple data exports.

3.5.2 Reading and Parsing the .res Files

To efficiently process multiple .res files, a Python-based data parsing routine was implemented. Each file was read line by line, and the following steps were executed programmatically:

1. Header reading: The first line was parsed to determine the number of raw sensor lines (n).
2. Extraction of valid raw channels: The next n lines were read and split by commas. Only those rows with a quality flag of “1” were retained. Invalid readings (flag = 0) were discarded.
3. Metadata parsing: The metadata line was split by commas to extract individual fields. The date and time fields were reconstructed into a unified timestamp (YYYY-MM-DD HH:MM). The year field (“25”) was automatically converted to “2025,” assuming 21st-century measurements.
4. Coordinate validation: Latitude and longitude were checked to ensure they fell within expected ranges for the Ostpark area (latitude 50.10–50.15, longitude 8.70–8.75).

5. Record assembly: All extracted fields were compiled into a structured table (DataFrame) containing:
- **BaumID**
 - **Timestamp (UTC)**
 - **Day of Year (DOY)**
 - **Species (Genus + species epithet)**
 - **Latitude / Longitude**
 - **Measurement status**
 - **Benchmark calibration group**
 - **Device file reference**
 - **Count of valid sensor channels**
 - **Record ID**

This process ensured that all .res files were parsed consistently, and any corrupted or incomplete files were automatically flagged for exclusion.

Final Output and Export (Section [3.5.1](#) and [3.5.2](#))

After parsing each Arborcheck .res file, numeric measurements are extracted, key metadata is captured, and a date string is derived from the filename. For every file, numbers are validated by a token, nonnumeric lines are treated as metadata, and missing fields are left blank. This yields a uniform schema.

Finally, records are assembled into one table and written to a CSV (`/content/sample_data/all_res_combined.csv`) with a header row. The header includes Date, BaumID, BaumArt, and named value columns (**Efficiency, Chlorophyll, Si1–Si4**). Variable-length rows are padded to match the widest.

A quick inspection confirms consistent columns and stable date formatting (DD/MM/YYYY). The file is the input for subsequent cleaning, categorisation, and modelling within the pipeline.

3.5.3 Cleaning and Transformation of Arborcheck Data

In this stage, the processed Arborcheck readings that were previously merged and exported as a CSV file (`/content/sample_data/all_res_combined.csv`) are further refined through a data cleaning and transformation procedure. The purpose of this step is to transform the raw numerical outputs of the Arborcheck measurements into meaningful and interpretable categorical indicators that can later be used for statistical analysis and machine learning modelling. The cleaned and categorised dataset is then exported as a new file named `/content/sample_data/Arborcheck_with_categories.csv`.

The Arborcheck device provides several sensor-derived variables, such as Efficiency, Chlorophyll, and four stress indicators (Si1 to Si4). These continuous numerical variables describe the physiological condition of the tree but are not directly interpretable for

management or comparison purposes. Therefore, this step introduces two new categorical fields `VitalityCategory` and `StressCategory` which summarise the overall vitality and stress level of each tree in standardised descriptive terms (e.g., “Good”, “Slight reduction”, “Severe”, etc.).

Before performing the transformation, the code checks and corrects possible inconsistencies in the dataset. For example, a column name such as "Genus " (with an unintended space) is automatically renamed to "Genus". This ensures that the dataset remains consistent and avoids column mismatches during later data joins or analyses.

Vitality Classification

The first transformation involves assessing the vitality status of each tree using two parameters: Efficiency and Chlorophyll content. These two measurements reflect how effectively the tree’s photosynthetic system is functioning.

The algorithm defines a function **VitalityCategory(efficiency, chlorophyll)** which assigns each record to one of four vitality levels:

- Critical reduction: when the Efficiency value is lower than -4.0
- Significant reduction: when Efficiency is between -4.0 and -3.0
- Slight reduction: when Efficiency is between -3.0 and -2.0
- Good: when Efficiency is above -2.0

If a reading for *Efficiency* is missing, the function returns “**None**” to prevent misclassification.

To refine this classification, Chlorophyll is also considered as a secondary criterion. Even if a tree’s Efficiency appears “**Good**”, a low **Chlorophyll value (below -2.5)** triggers a downgrade to “**Slight reduction**”, as this indicates early pigment loss and potential physiological stress. This rule prevents overestimating tree health when chlorophyll degradation already signals reduced vitality.

Overall, this function combines two physiological indicators into a single, interpretable vitality class that can be used for visualisation, comparison between species, or further modelling.

In later stages of model development, the two lowest vitality levels Critical reduction ($\text{Efficiency} < -4.0$) and Significant reduction ($-4.0 \leq \text{Efficiency} < -3.0$) were merged into a single “Critical reduction” category. This adjustment was made to reduce class imbalance, as the number of observations in these two categories was comparatively small. When four classes were used, the model performance was unsatisfactory due to the imbalance; however, after merging the classes, the classification results improved considerably.

Stress Classification

The second transformation focuses on assessing stress levels, using four indices (Si1, Si2, Si3, and Si4). These indices describe different spectral or physiological responses of the trees under stress conditions.

The process involves two functions: **_si_band()** and **stress_category()**. The first function, **_si_band()**, translates each continuous Si value into a band or severity level (1 to 3). This is done according to pre-defined thresholds that differ slightly for Si1 compared to the other indices, since Si1 is directionally sensitive (only negative deviations indicate stress). For example, large negative Si1 values below -2.5 are assigned the highest severity band (3). For Si2–Si4, the classification is symmetric, so both large positive and negative deviations indicate higher stress magnitudes.

Once all four indices are assigned their individual bands, the main function **stress_category()** combines them to determine an overall stress level for the tree. The procedure works as follows:

1. Identify the worst (highest) band among all four indices – this represents the strongest stress signal observed.
2. Calculate the average of all four indices (ignoring missing values).
3. Based on these two quantities, assign the final StressCategory as:
 - a. Severe: if any index shows a very strong deviation (band 3)
 - b. Moderate: if the worst band is 2 and the average magnitude is also relatively high (≥ 1.5)
 - c. Mild: for smaller deviations
 - d. No Stress: when all indices are within normal ranges or when values are missing

This approach follows a conservative principle, meaning that even a single severely stressed signal is enough to classify the tree as “Severe”. The average value ensures that consistent moderate deviations across indices are not underestimated, while small, isolated fluctuations are treated as “Mild” or “No Stress”.

Final Output and Export (Section [3.5.3](#))

After computing both the **VitalityCategory** and **StressCategory** for each record, these new columns are appended to the dataset. The final version of the table thus includes both the raw sensor measurements and their categorical interpretations.

Finally, the enriched dataset is exported to a new CSV file (**Arborcheck_with_categories.csv**) without an index column. This cleaned and transformed dataset serves as the input for subsequent analytical and machine learning procedures, where categorical vitality and stress labels are used as target variables for model training and evaluation.

3.5.4 Merging with PlanetScope Satellite Data

After completing the transformation and categorization of the Arborcheck dataset, it was integrated with the PlanetScope spectral reflectance dataset prepared in Section [3.4.8](#).

This step connected ground-measured physiological data to satellite-observed spectral characteristics.

The merging process was executed as follows:

1. **Key Matching:** Both datasets contained the same BaumID identifier. Records were merged based on this ID and the date of measurement. Since satellite imagery and field measurements were collected within the same time frame, a ± 1 -day temporal tolerance was used to account for potential differences between measurement and satellite acquisition times.
2. **Selection of High-Quality Scenes:** In cases where multiple PlanetScope images were available for the same tree and date, only the image with the highest clear-sky confidence ($\geq 80\%$) was retained. This ensured that the reflectance values used were free from cloud or haze interference.
3. **Data Validation:** The merged dataset was manually validated by cross-checking sample entries for consistency between tree IDs, coordinates, and acquisition dates.

The resulting dataset contained:

- Tree metadata (**BaumID, Species, Date, Latitude, Longitude**)
- Physiological measurements (Fv/Fm, SI1–SI4, Vitality Category and Stress Category)
- Spectral reflectance values (Bands B1–B8)

This merged dataset represented the complete integration of field and satellite data, providing a robust foundation for model development.

3.5.5 Final Output and Model Preparation

The merged dataset was exported as a comma-separated file (Merged_Field_Satellite.csv). Each row corresponded to a single tree observation that had both valid physiological measurements and a matching satellite observation.

4 Machine Learning Model Development

The development of the machine learning (ML) models for this study was preceded by an extensive process of data preparation and integration that aimed to combine high-resolution remote sensing imagery with in-situ physiological measurements. This section details the complete workflow, from the delineation of the study plots and image collection to data preprocessing, temporal alignment, and dataset consolidation prior to model training.

Please find all scripts in GitHub – <https://github.com/santu001/urban-tree-stress.git>

4.1 Data Collection, Preparation and Integration

4.1.1 Study Tree Selection and Region of Interest Delineation

The study was conducted in collaboration with the Grünflächenamt (Municipal Green Space Office) of the City of Frankfurt am Main, which provided both logistical support and access to field data. Based on a joint consultation, eight individual trees were selected within the Ostpark area as representative subjects for detailed analysis. The trees were chosen to represent a range of species and vitality conditions typical of Frankfurt's urban forest composition.

To ensure precise spatial correspondence between field data and satellite imagery, two Regions of Interest (ROIs) were created in Google Earth Engine (GEE), each covering a portion of Ostpark. The polygons were drawn manually around the relevant field sites using the satellite base map for spatial guidance ([Figure 3.10](#)). Each ROI was saved as a geometry variable, with an associated name and image collection definition.

The ROIs were defined as follows:

- Frankfurt_Ost_1_Bbox (Variable: geo_ROI_BBox)
 - Short name: FRA_1
 - Image Collection name: GDE_8B_FRA_PSScene1
 - Contains four trees (BaumIDs: 801, 808, 809, 749)
- Frankfurt_Ost_2_Bbox (Variable: geo_ROI_BBox)
 - Short name: FRA_2
 - Image Collection name: GDE_8B_FRA_PSScene2
 - Contains four trees (BaumIDs: 444, 578, 571, 569)

Each Bbox represented a distinct image collection area. Together, these two Bboxes encompassed the entire set of eight trees used in the 2024 data collection campaign.

4.1.2 Ground Data Collection and Physiological Measurements

Field data were collected between 15 August and 15 September 2024 using the Arborcheck physiological assessment system, which provides leaf-level measurements of tree vitality. The field campaign was carefully scheduled to coincide with PlanetScope satellite overpass dates to ensure the temporal synchronization of satellite and ground datasets.

For each of the eight trees, the following parameters were recorded:

- **Dark-adapted chlorophyll fluorescence efficiency (Fv/Fm):** representing the maximum quantum efficiency of photosystem II (PSII) and indicating the photosynthetic performance of the leaf.
- **Leaf chlorophyll content index:** representing pigment concentration and nutrient status.
- **Stress Indices S1–S4:** four numerical parameters derived from Arborcheck’s physiological models, representing distinct stress response pathways (e.g., photochemical limitation, pigment degradation, and metabolic inhibition).

These field measurements were exported in tabular form and served as the ground truth for the machine learning model labels, with each observation categorized into one of three physiological classes: Healthy, Drought Stress, or Nutrient Stress.

4.1.3 Satellite Data Acquisition

The remote sensing data were obtained from PlanetScope’s 8-band SuperDove constellation, which provides daily surface reflectance imagery at approximately 3–5 m spatial resolution. The eight bands (Coastal Blue, Blue, Green I, Green, Yellow, Red, Red-edge, and Near-Infrared) were used as the primary features for spectral characterization.

For each ROI, a dedicated image collection was defined in GEE using the asset names:

- GDE_8B_FRA_PSScene1 for Frankfurt_Ost_1_Bbox
- GDE_8B_FRA_PSScene2 for Frankfurt_Ost_2_Bbox

The date range for the imagery was restricted to the period of 15 August to 15 September 2024, corresponding exactly with the field measurement campaign. All images within this window were downloaded and processed following a standardized workflow (detailed below).

4.1.4 Image Preprocessing and Quality Control

A rigorous preprocessing pipeline was implemented to ensure the radiometric and geometric integrity of the PlanetScope images. The following steps were applied within GEE and during post-processing:

- 1) **Cloud and shadow masking:** Each image was filtered using Planet’s quality assurance (QA) bands to remove pixels affected by cloud cover, haze, or cast shadows.
- 2) **Radiometric correction:** All scenes were converted to surface reflectance units (0–1) to standardize illumination and atmospheric conditions.
- 3) **Spatial subsetting:** Each image was clipped to its respective ROI (Frankfurt_Ost_1_Bbox or Frankfurt_Ost_2_Bbox) to limit the dataset to the relevant area.
- 4) **Temporal filtering:** Only cloud-free or minimally contaminated images were retained for analysis.

Due to the high frequency of cloudy days during late summer, many images were excluded, leaving eight valid acquisition dates that matched the field data collection period.

4.1.5 Per-Tree Data Extraction and Synchronization

The next step involved linking each tree’s field measurements with the corresponding satellite-derived spectral information. To achieve this, each tree canopy was manually delineated as a polygon in GEE ([Figure 3.10](#)). These geometries were converted into FeatureCollections, with each feature containing metadata attributes (FID and BaumID).

For each acquisition date, the mean reflectance values across the eight PlanetScope spectral bands were extracted for every tree polygon. The temporal matching was then performed based on the acquisition dates of the images and the timestamps of the Arborcheck field measurements, using a tolerance of ± 1 day.

After synchronization and filtering, the resulting dataset comprised 57 valid observations, representing eight trees across eight temporal acquisitions. Each record included both the remote sensing features (spectral bands) and the corresponding physiological measurements.

4.1.6 Dataset Augmentation Using Historical Data (2022–2023)

Initial testing of the machine learning models using this dataset produced unsatisfactory performance metrics, mainly due to the limited number of samples and the restricted temporal range. To address this, the Grünflächenamt provided access to historical Arborcheck data collected during 2022 and 2023, recorded in the different part of Frankfurt region.

A subset of four additional trees was selected from these archived datasets to expand the temporal coverage and increase the diversity of physiological states represented. These trees were digitized and georeferenced in GEE, forming a third ROI titled Frankfurt_Ost_3_Bbox.

The associated image collection was defined as:

- Frankfurt_Ost_3_Bbox $\rightarrow$ GDE_8B_FRA_PSScene3

For this new region, the same standardized workflow was applied:

- 1) Creation of tree polygons and feature attributes (FID and BaumID).
- 2) Download of PlanetScope imagery for the corresponding 2022–2023 measurement periods.
- 3) Application of cloud masking and radiometric correction.
- 4) Extraction of tree-level mean reflectance values.

This process yielded approximately 30 additional data points, representing the four newly included trees (BaumIDs 154, 161, 152, and 153). These were merged with the 2024 dataset to produce an expanded dataset with ~87 tree-level records in total.

4.1.7 Data Integration and Final Feature Set

The merged dataset included features derived from all three PlanetScope image collections (PSScene1, PSScene2, and PSScene3) and corresponding Arborcheck field data. Each record contained:

- **Eight spectral bands:** Coastal Blue, Blue, Green I, Green, Yellow, Red, Red-Edge, and NIR.
- **Physiological metrics:** Fv/Fm, chlorophyll content index, and stress indicators (S1–S4).
- **Class label:** categorical stress type (Vitality Category, StressCategory).

All features were normalized to a common scale and checked for multicollinearity using Pearson correlation analysis prior to model training. Missing data were handled by listwise deletion to preserve data integrity.

This consolidated, high-quality dataset formed the foundation for the machine learning model development. The subsequent section ([Section 4.2](#)) details the training, optimization, and evaluation of three classifiers Logistic Regression, Support Vector Machine (SVM), and Random Forest which were designed to predict the physiological stress class of each tree based on the integrated remote sensing and ground data.

4.2 Feature Engineering

Feature engineering represents the most critical preparatory stage in the machine learning workflow, as it transforms raw satellite reflectance into structured numerical features that can meaningfully describe vegetation condition. For this study, all models (Logistic Regression, Support Vector Machine, and Random Forest) were trained using a common set of predictor

variables derived from PlanetScope's 8-band multispectral data, with only minor adjustments between models.

The goal of this feature-engineering process was to extract biophysically interpretable variables that capture changes in pigment concentration, photosynthetic efficiency, canopy structure, and general vitality of individual trees.

4.2.1 Spectral Inputs: PlanetScope 8-Band System

Each tree's reflectance was extracted from the PlanetScope SuperDove constellation, which captures eight narrow multispectral bands at ~3–5 m spatial resolution. These bands collectively cover the visible, red-edge, and near-infrared (NIR) portions of the electromagnetic spectrum, making them highly suitable for vegetation monitoring.

PlanetScope 8-band SuperDove (PSB.SD): what each band captures and why it matters

1) Coastal Blue (443 nm; FWHM $\approx$ 20 nm) B1:

Ultra-blue wavelengths penetrate clear, shallow water and are sensitive to aerosols and fine particulates. Use for coastal/bathymetric mapping, water quality (suspended sediment, chlorophyll-a proxies), haze detection, and cross-sensor aerosol correction. Interoperable with Sentinel-2 B1. [\[50\]](#)

2) Blue (490 nm; ~50 nm) B2:

Captures water colour and vegetation reflectance on the blue shoulder of the chlorophyll absorption. Useful for turbidity, shallow-water depth cues, atmospheric correction, and as a "blue" channel in true-colour composites. Interoperable with Sentinel-2 B2. [\[B2\]](#)

3) Green I (531 nm; ~36 nm) B3:

Narrow green band centered near chlorophyll-b absorption; enhances separability of vegetation species and stress compared to a single broad green band. Valuable for vegetation indices that exploit the green peak and for reducing confusion between vegetation and certain artificial materials. (No direct Sentinel-2 equivalent.). [\[50\]](#)

4) Green (565 nm; ~36 nm) B4:

Covers the classic green reflectance peak of healthy vegetation. Supports vigor mapping, general land cover, and green-based indices (e.g., $GNDVI = (NIR - Green) / (NIR + Green)$). Interoperable with Sentinel-2 B3. [\[50\]](#)

5) Yellow (610 nm; ~20 nm) B5:

Bridges the green and red absorption regions; useful for pigment-sensitive work (carotenoids/anthocyanins), crop senescence mapping, and fine-grained material/soil discrimination when paired with red and red-edge. (No Sentinel-2 equivalent.) [\[50\]](#)

6) Red (665 nm; ~31 nm) B6:

Strong chlorophyll-a absorption region; foundational for vegetation vigor/stress metrics (NDVI, EVI2) and for delineating built-up versus vegetated surfaces. Interoperable with Sentinel-2 B4. [\[50\]](#)

7) Red-edge (705 nm; ~15 nm) B7:

Sits on the steep slope between red absorption and NIR plateau; highly sensitive to chlorophyll content, leaf structure, and early stress before changes appear in red/NIR. Enables red-edge indices (e.g., NDRE, chlorophyll indices) vital for precision agriculture and forest health monitoring. Interoperable with Sentinel-2 B5. [\[50\]](#)

8) Near-infrared (865 nm; ~40 nm) B8:

Captures the vegetation “plateau” driven by internal leaf scattering; correlates with biomass, canopy structure, and LAI. Core input for NDVI/GNDVI/NDRE and false-color (NIR-Red-Green) composites. Interoperable with Sentinel-2 B8a. [\[50\]](#)

These eight bands together provide complementary information – the visible region (B1–B5) reflects pigment absorption and photosynthetic activity, while the red-edge (B7) and NIR (B8) respond to structural and biochemical changes within leaves and canopies

4.2.2 Derived Spectral Features

Beyond raw reflectance, a series of derived indices were computed to amplify subtle spectral relationships. These indices help normalize the effects of illumination, viewing geometry, and soil background while highlighting physiological processes relevant to vegetation stress.

Each index represents a different biophysical or biochemical property, making them valuable for multi-dimensional learning in the models.

(a) Greenness and Canopy Structure

- **NDVI = (B8 – B6) / (B8 + B6)**

The Normalized Difference Vegetation Index remains the most widely used indicator of canopy greenness and overall vegetation vigor. It exploits the contrast between high NIR reflectance (healthy canopy) and strong red absorption (chlorophyll activity). NDVI values increase with chlorophyll density and photosynthetically active biomass.

In model training, NDVI helps establish a baseline separation between healthy and stressed vegetation.

- **$GNDVI = (B8 - B4) / (B8 + B4)$**

The Green NDVI substitutes the red band with the green band, enhancing sensitivity to chlorophyll variation in moderate or dense canopies where NDVI often saturates. It improves discrimination of subtle chlorophyll losses associated with nutrient deficiency or early drought.

(b) Chlorophyll and Nutrient Sensitivity

- **$NDRE = (B8 - B7) / (B8 + B7)$**

The Normalized Difference Red-Edge Index isolates the red-edge region, which shifts toward shorter wavelengths under chlorophyll loss. NDRE is thus a direct proxy for chlorophyll and nitrogen content and provides a sensitive indicator for pre-visual stress detection.

- **$CLRE = (B8 / B7) - 1$ and $CIgreen = (B8 / B4) - 1$**

The Chlorophyll Indices compare NIR reflectance to red-edge or green reflectance. These indices increase with leaf chlorophyll concentration and canopy thickness, allowing the models to capture longer-term pigment dynamics.

- **$MTCI = (B8 - B7) / (B7 - B6)$**

The MERIS Terrestrial Chlorophyll Index extends sensitivity to a wider chlorophyll range. By using both red and red-edge regions, it can detect saturation points and differentiate between moderate and high pigment levels.

(c) Photochemical and Short-Term Stress Indicators

- **$PRI = (B3 - B4) / (B3 + B4)$**

The Photochemical Reflectance Index measures rapid adjustments in the xanthophyll cycle, which regulates excess light energy in photosystems. A lower PRI indicates reduced photosynthetic efficiency or photoinhibition due to stress (e.g., heat, drought). Incorporating PRI enhances the model's ability to capture short-term physiological fluctuations not visible in NDVI.

(d) Soil and Structural Adjustment Indices

- **$EVI = 2.5 \times (B8 - B6) / (B8 + 6 \times B6 - 7.5 \times B2 + 1)$**

The Enhanced Vegetation Index corrects for atmospheric and soil background noise, improving linearity in dense canopies. It helps ensure that reflectance differences are due to physiological factors rather than illumination or background variations.

- **SAVI = $((B8 - B6) \times (1 + L)) / (B8 + B6 + L)$ with $L = 0.5$**

The Soil-Adjusted Vegetation Index reduces soil reflectance contamination, relevant for urban environments where mixed pixels often include grass, soil, or pavement.

- **MSAVI2 = $0.5 \times (2 \times B8 + 1 - \sqrt{((2 \times B8 + 1)^2 - 8 \times (B8 - B6))})$**

The Modified SAVI automatically adjusts for soil brightness and vegetation density. Including this index stabilizes predictions across heterogeneous surfaces like park edges or sparse canopy zones.

- **RDVI = $(B8 - B6) / \sqrt{(B8 + B6)}$**

The Renormalized Difference Vegetation Index linearizes NDVI's response, helping the model retain discriminatory power even at high chlorophyll levels.

(e) Ratio-Based Reflectance Contrasts

- **Ratio_RE_Red = $B7 / B6$**

Quantifies the transition from chlorophyll absorption (red) to reflectance (red edge). It is sensitive to pigment degradation and canopy optical depth.

- **Ratio_Red_NIR = $B6 / B8$**

Serves as an inverse of greenness indices. Higher values may indicate stress, senescence, or leaf thinning.

- **Ratio_Blue_Red = $B2 / B6$**

Compares blue to red reflectance; increases in this ratio often signal loss of chlorophyll and pigment bleaching under nutrient stress.

- **VARI = $(B4 - B6) / (B4 + B6 - B2)$**

The Visible Atmospherically Resistant Index improves color balance and helps the model learn visual contrasts under variable lighting. It supports discrimination between shadowed and sunlit canopies.

(f) Water and Moisture Contrast

- **NDWI = $(B4 - B8) / (B4 + B8)$**

Here the Normalized Difference Water Index uses the visible–NIR formulation of McFeeters (originally designed for water detection). Although PlanetScope lacks SWIR

bands, this ratio still captures general canopy wetness and brightness contrasts, indirectly aiding discrimination between dry and hydrated canopies.

Each of these features contributes complementary spectral information. Together, they help the machine learning models detect both long-term biochemical changes (e.g., pigment or nutrient depletion) and short-term physiological responses (e.g., photoinhibition, reduced light-use efficiency).

4.2.3 Transformations and Scaling

To improve numerical stability and linear separability for the statistical models, several feature transformations were applied:

- **Logarithmic scaling:** For all eight raw bands, logarithmic transforms ($\log_{10}(B1) \dots \log_{10}(B8)$) were computed. This transformation compresses large reflectance values, reducing skewness and ensuring that the logistic regression and SVM models both of which assume linear relationships can better capture proportional differences between stress states.
- **Standardization:** For models that rely on gradient-based optimization (Logistic Regression and SVM), features were standardized to zero mean and unit variance using `StandardScaler`. This ensures that all features contribute equally to the cost function and avoids dominance by high-magnitude variables.
- **Random Forest exception:** Tree-based methods such as Random Forest are scale-invariant and unaffected by monotonic transformations. Therefore, the $\log_{10}$ features were excluded from the Random Forest pipeline, and raw reflectance and index features were used directly.

4.2.4 Data Cleaning and Quality Control

To maintain feature reliability and prevent artifacts in model training:

- 1) All feature computations included a small constant ($\epsilon = 1 \times 10^{-9}$) in denominators to prevent division-by-zero errors.
- 2) Any non-finite values (NaN, $+\infty$, $-\infty$) resulting from cloud contamination or data gaps were removed via listwise deletion.
- 3) Each feature was visually inspected for range consistency and physically plausible distribution.
- 4) Finally, feature correlation matrices were reviewed to ensure that redundant variables did not dominate the model's learning process.

In addition to these preprocessing steps, class structure adjustments were made to improve data balance during model training, which mentioned in [Section 3.5.3](#) too. Specifically, the two lowest vitality categories Critical reduction ($\text{Efficiency} < -4.0$) and Significant reduction ($-4.0 \leq \text{Efficiency} < -3.0$) were merged into a single Critical reduction class. This modification effectively mitigated class imbalance issues observed in the initial four-class setup and led to improved overall model performance.

4.2.5 Final Feature Composition

After cleaning and validation, each tree–date observation contributed:

- 8 raw reflectance bands (B1–B8)
- 16 derived indices capturing chlorophyll, structure, photochemical, and pigment dynamics
- 8 log-transformed bands (used only in Logistic Regression and SVM models)

This produced a total of 32 predictor variables for the Logistic Regression and SVM models, and 24 predictors for the Random Forest model.

The combined feature matrix provided a robust multidimensional representation of canopy condition, linking optical, physiological, and structural properties of the trees to the corresponding Arborcheck field measurements. By integrating complementary spectral sensitivities, the engineered features enabled the machine learning algorithms to learn complex, nonlinear relationships between spectral reflectance and physiological stress states.

4.3 Logistic Regression Model Development

Overview

The first classification framework developed for this research employed a multinomial Logistic Regression algorithm, implemented in Python using the scikit-learn machine learning library.

This model served as an interpretable baseline to classify both tree vitality categories and stress categories based on a merged dataset of satellite-derived reflectance features and field physiological observations (Arborcheck data).

Logistic Regression was selected primarily for its interpretability, simplicity, and ability to provide direct probabilistic outputs. These characteristics make it especially suitable for small datasets where transparency and feature influence are of interest.

4.3.1 Model Objective and Rationale

The goal of this model was to predict two ecological indicators:

1. **Vitality Category** – an index of overall physiological tree condition.
2. **Stress Category**– a classification reflecting the predominant type of physiological stress (e.g., drought-related, nutrient-related, or healthy).

The logistic regression model estimates the probability that an observation belongs to a given class, using a linear combination of spectral features. Mathematically, it models the log-odds of class membership as:

$$\log (P(y = c) / P(y = \text{reference})) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

where y is the target class, x_i are the input features (reflectance or spectral indices), and β_i are the model coefficients estimated through maximum likelihood optimization.

Each coefficient thus indicates the strength and direction of association between a predictor and a specific vitality or stress class.

4.3.2 Data Preparation

The combined input dataset contained pre-calculated features derived from PlanetScope bands and vegetation indices. The two classification targets were specified as:

```
TARGETS = ["VitalityCategory", "StressCategory"]
```

Data were imported using a flexible function that accepts both Excel and CSV formats:

```
def load_table(path: str) -> pd.DataFrame:
    ext = Path(path).suffix.lower()
    if ext in [".xlsx", ".xls"]:
        return pd.read_excel(path)
    return pd.read_csv(path)
```

Rows containing missing target values were excluded to ensure consistency in training and testing.

The dataset was divided into a training subset (75%) and a test subset (25%) using stratified sampling, which preserves class proportions across splits:

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=TEST_SIZE, random_state=RANDOM_STATE, stratify=y
)
```

This design was critical to maintain representativeness across different vitality and stress levels, particularly given the moderate class imbalance in the available data.

4.3.3 Model Architecture and Implementation

The Logistic Regression classifier was encapsulated within a scikit-learn pipeline, which combined standardization and model fitting into a single reusable structure. This design ensures that the same pre-processing steps are consistently applied during both training and prediction phases.

```
def make_model() -> Pipeline:
    """StandardScaler -> LogisticRegression (multinomial, class-
balanced)"""
    clf = Pipeline([
        ("scaler", StandardScaler()),
        ("logreg", LogisticRegression(
            multi_class="multinomial",
            solver="saga",
            class_weight="balanced",
            C=1.0,
            max_iter=5000,
            random_state=RANDOM_STATE,
            n_jobs=-1
        ))
    ])
    return clf
```

Component Details

- 1) **StandardScaler:** Normalizes each feature to have zero mean and unit variance.

This step ensures that all spectral and index-based predictors contribute proportionally, preventing dominance by variables with large numeric scales.

- 2) **Multinomial Logistic Regression** Configured with the following parameters:

- `multi_class="multinomial"`: activates the softmax function to handle multi-class classification directly.
- `solver="saga"`: efficient and robust optimization algorithm suitable for high-dimensional, class-weighted problems.
- `class_weight="balanced"`: automatically reweights classes inversely to their frequency, addressing imbalance between healthy and stressed samples.
- `C=1.0`: controls the inverse of regularization strength (moderate penalization to reduce overfitting).
- `max_iter=5000`: increases convergence iterations to ensure stable optimization.
- `random_state=42`: guarantees reproducibility of results.
- `n_jobs=-1`: enables parallel computation to accelerate training.

This parameterization was designed to maximize stability, interpretability, and fairness across all classes in a relatively small dataset.

4.3.4 Model Training and Evaluation Procedure

The function `fit_eval_one_target()` automated the training, prediction, and evaluation workflow for each classification target.

```
def fit_eval_one_target(X: pd.DataFrame, y_series: pd.Series,
                        target_name: str):
    le = LabelEncoder()
    y = le.fit_transform(y_series.astype(str))
    class_names = list(le.classes_)

    ...

    model = make_model()
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)
```

Step 1: Label Encoding

Before model fitting, string-based class labels (e.g., “Healthy”, “Drought”, “Nutrient Stress”) were transformed into numeric form using `LabelEncoder()`. This conversion allows the Logistic Regression algorithm to process categorical targets computationally.

Step 2: Model Fitting

The model was trained (`fit()`) using the training dataset.

During this phase, the optimization algorithm iteratively adjusted the model coefficients (β_i) to minimize the negative log-likelihood loss, ensuring the best separation between the predefined classes.

Step 3: Prediction and Evaluation

After training, the model predicted class labels on the unseen test set. Multiple evaluation metrics were calculated to measure generalization performance:

```
acc = accuracy_score(y_test, y_pred)
mfl = f1_score(y_test, y_pred, average="macro", zero_division=0)
bacc = balanced_accuracy_score(y_test, y_pred)
```

```
rep = classification_report(y_test, y_pred, target_names=class_names,
output_dict=True, zero_division=0)
cm = confusion_matrix(y_test, y_pred)
```

Performance metrics:

- **Accuracy:** Proportion of all correctly predicted samples.
- **Balanced Accuracy:** Mean recall across all classes, ensuring equal weight to minority classes.
- **Macro F1-score:** Average of F1-scores per class, combining both precision and recall.
- **Classification Report:** A detailed summary of per-class precision, recall, and F1 values.
- **Confusion Matrix:** Quantitative visualization of correct vs. misclassified samples.

This standardized evaluation framework ensures comparability with subsequent Support Vector Machine (SVM) and Random Forest models developed later in the study.

4.3.5 Visualization of Model Results

To complement numerical metrics, the model generates a suite of graphical outputs to support interpretability and thesis presentation.

All figures are saved automatically under `logreg_outputs/plots_<target_name>/.`

- 1) **Confusion Matrix Plot:** A heatmap visualization showing agreement between true and predicted labels. It helps identify which classes are most frequently confused.
- 2) **Feature Importance Plot:** Derived from the average absolute values of standardized regression coefficients. The top 20 predictors are displayed in descending order, revealing which spectral indices most strongly influence model predictions.
- 3) **Per-Class Performance Plot:** Bar plots visualizing precision, recall, and F1-score for each class, enabling quick comparison of how the model performs across vitality or stress levels.

Each plot is automatically saved in .png format with high resolution (300 dpi) to ensure inclusion quality for reports and publications.

4.3.6 Execution Flow

The complete process is orchestrated by the `main()` function, which executes sequentially:

- 1) Data loading and validation
- 2) Feature computation (excluded here, detailed separately)
- 3) Model training and evaluation for both targets

4) Result export and visualization

The script executes automatically through:

```
if __name__ == "__main__":  
    main()
```

This structure ensures reproducibility and automation, allowing straightforward reruns on updated datasets.

4.3.7 Summary and Interpretation

The multinomial Logistic Regression model provided a transparent and interpretable baseline for classifying tree vitality and stress conditions.

Through the integration of standardized scaling, class balancing, and robust regularization, it was able to handle a small, imbalanced dataset effectively while maintaining interpretability.

Although logistic regression is a linear model and may not capture complex nonlinear relationships between spectral features and physiological stress responses, it serves as an essential benchmark for understanding which features most strongly explain variation in vitality and stress outcomes.

Its results also formed the foundation for comparison with more complex machine learning models, such as Support Vector Machines and Random Forests, discussed in subsequent sections.

4.4 Support Vector Machine Model Development

Overview

The second classifier developed in this study is a Linear Support Vector Machine (SVM) configured for multi-class prediction and probability calibration. The implementation uses scikit-learn and trains two independent models targeting (i) VitalityCategory and (ii) StressCategory. Compared to Logistic Regression, the linear SVM emphasizes maximizing the margin between classes and is often competitive on small to medium datasets with high-dimensional features. Because standard linear SVMs do not produce probabilistic outputs, we wrap the base classifier with CalibratedClassifierCV to obtain reliable class probability estimates for interpretability and downstream analysis.

4.4.1 Model Objective and Rationale

The model predicts:

- 1) **VitalityCategory** - overall physiological condition.
- 2) **StressCategory** - dominant stress condition.

A linear SVM seeks a set of hyperplanes that maximizes the margin between classes in feature space. For the multi-class setting, scikit-learn's LinearSVC uses a one-vs-rest strategy under the hood: it fits one linear decision function per class against all others. This choice is appropriate when:

- the number of features is moderate to high (here: raw bands, indices, and log-transformed bands),
- sample size is limited, and
- a linear boundary is a reasonable first approximation (improving interpretability and computation).

Because LinearSVC returns decision scores rather than probabilities, probability calibration (Platt scaling via `method="sigmoid"`) is applied with cross-validation. This yields well-calibrated probabilities that help interpret prediction confidence and support threshold-based decisions.

4.4.2 Model Objective and Rationale

The pipeline operates on a pre-processed table that includes all necessary features (raw bands, indices, log transforms) and the two targets:

```
TARGETS = ["VitalityCategory", "StressCategory"]
```

Files are loaded via a small helper that supports Excel and CSV:

```
def load_table(path: str) -> pd.DataFrame:
    ext = Path(path).suffix.lower()
    return pd.read_excel(path) if ext in [".xlsx", ".xls"] else
pd.read_csv(path)
```

Rows with missing target labels are removed. Data are then split into train (75%) and test (25%) using stratified sampling to preserve class balance:

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=TEST_SIZE, random_state=RANDOM_STATE,
    stratify=y)
```

Stratification is essential because both vitality and stress classes are **imbalanced** (some stress types are rarer).

4.4.3 Model Architecture and Implementation

The SVM is defined inside a scikit-learn Pipeline that couples scaling with a calibrated linear classifier:

```
from sklearn.svm import LinearSVC
from sklearn.calibration import CalibratedClassifierCV
from sklearn.pipeline import Pipeline

def make_model() -> Pipeline:
    base = LinearSVC(C=1.0, class_weight="balanced",
                    random_state=RANDOM_STATE)
    cal = CalibratedClassifierCV(base, cv=3, method="sigmoid")
    return Pipeline([("scaler", StandardScaler()), ("cal", cal)])
```

Component Details and Design Choices

1. **StandardScaler:** SVMs are scale-sensitive: features with larger magnitudes can dominate the margin. Standardization (zero mean, unit variance) ensures that each feature contributes proportionally.
2. **LinearSVC (base estimator)**
 - a. Linear kernel: promotes interpretability (coefficients map directly to features) and is efficient for moderately high-dimensional data.
 - b. $C=1.0$: regularization strength; balances margin maximization and training error. With limited data, a moderate C avoids overfitting.
 - c. `class_weight="balanced"`: compensates for class imbalance by weighting classes inversely to their frequency.
 - d. `random_state=RANDOM_STATE`: reproducibility.
3. **CalibratedClassifierCV**
 - a. `cv=3` and `method="sigmoid"` (Platt scaling): fits an internal cross-validated calibration model that maps decision scores to probabilities.
 - b. Provides probability estimates per class – valuable for interpretability, threshold tuning, and communicating uncertainty in ecological decision-making.

This design yields a margin-based, class-balanced, and probability-calibrated linear model that is robust on small, imbalanced datasets.

4.4.4 Training and Evaluation Workflow

All steps for a single target are encapsulated in `fit_eval_one_target()`:

```
def fit_eval_one_target(X, y_series, target_name):
    le = LabelEncoder()
    y = le.fit_transform(y_series.astype(str))
```

```
...  
model = make_model()  
model.fit(X_train, y_train)  
y_pred = model.predict(X_test)
```

Step 1: Label Encoding: Categorical class strings are converted to integers via `LabelEncoder`, and the encoder is saved for consistent decoding later.

Step 2: Model Fitting: The pipeline (scaler → calibrated `LinearSVC`) is fit on the training set. Internally, SVM hyperplanes are learned; then calibration is performed via 3-fold CV to map decision scores to probabilities.

Step 3: Prediction and Metrics: On the test set (25% holdout), predictions are evaluated with complementary metrics:

```
acc = accuracy_score(y_test, y_pred)  
mfl = f1_score(y_test, y_pred, average="macro")  
bacc = balanced_accuracy_score(y_test, y_pred)  
rep = classification_report(y_test, y_pred, target_names=class_names,  
zero_division=0)  
cm = confusion_matrix(y_test, y_pred)
```

- Accuracy – overall correctness.
- Balanced Accuracy – average recall across classes; guards against dominance by majority class.
- Macro F1 – equal-weighted harmonic mean of precision and recall across classes.
- Classification Report – per-class precision, recall, F1; useful for identifying weak classes.
- Confusion Matrix – visualizes misclassification patterns.

4.4.5 Visualization and Interpretability

To complement numerical metrics, the model generates a suite of graphical outputs to support interpretability and thesis presentation.

- 1) **Confusion Matrix:** Heatmap of true vs. predicted classes to spot systematic confusions.
- 2) **Top-20 Feature Importance (|coef|):** Although the probability model is calibrated; we extract linear coefficients from the base SVM to gauge feature influence:

```
coefs = model.named_steps["cal"].base_estimator_.coef_  
importance = np.mean(np.abs(coefs), axis=0)
```

The mean absolute coefficient across one-vs-rest classifiers serves as a proxy importance. The top 20 are plotted as a horizontal bar chart for readability.

- 3) **Per-Class Performance (precision, recall, F1):** Bar plot from the classification report; highlights which classes need more data or different features.

4.4.6 Execution Flow

The `main()` function orchestrates the pipeline:

- 1) Load data and validate required columns (bands + targets).
- 2) Drop missing targets; build features (details covered elsewhere).
- 3) For each target in ["VitalityCategory", "StressCategory"]:
 - a. fit/evaluate (`fit_eval_one_target`),
 - b. save model & encoder,
 - c. export plots and holdout predictions.
- 4) Write summary CSV and export summary bar plot.

```
if __name__ == "__main__":  
    main ()
```

4.4.7 Summary and Interpretation

The calibrated Linear SVM offers a strong, margin-based baseline that is well-suited to the study's small, imbalanced dataset and moderately high-dimensional feature space. Three design choices are central:

- Linear kernel for computational efficiency and interpretability.
- Class balancing to avoid bias toward majority classes.
- Probability calibration to obtain reliable class probabilities, improving interpretability and enabling threshold tuning in management contexts.

While linear SVM's capture only linear decision boundaries, their calibrated probabilities and clear coefficient-based importance (proxy) make them an excellent complement to Logistic Regression. Together, these models provide transparent baselines before introducing more flexible non-linear methods (e.g., kernel SVM's or tree-based ensembles) in subsequent sections.

4.5 Random Forest Model Development with Extended Spectral and Textural Features

In this section, the development of two separate Random Forest (RF) models for the classification of Vitality Category and Stress Category is presented.

The models were trained using multispectral reflectance bands from PlanetScope imagery, augmented with a set of vegetation indices and optional texture-based features derived from local patch statistics.

The overall workflow is implemented in the Python script. This pipeline performs the following key steps:

- 1) Data loading and preprocessing
- 2) Feature engineering computation of raw band ratios, vegetation indices, and texture metrics
- 3) Model training separate RF models for Vitality and Stress
- 4) Evaluation and visualization
- 5) Artifact saving for inference

4.5.1 Data Input and Configuration

The training data were provided as an Excel file containing PlanetScope reflectance bands (B1–B8) along with categorical labels for Vitality and Stress.

Configuration parameters were defined at the beginning of the script:

```
DATA_PATH = r"/content/Data_To_Train_3_vitality.xlsx"
RAW_BANDS = ["B1", "B2", "B3", "B4", "B5", "B6", "B7", "B8"]
TARGET_VITALITY = "VitalityCategory"
TARGET_STRESS = "StressCategory"
TEST_SIZE = 0.2
RANDOM_STATE = 42
```

Eight PlanetScope bands (B1–B8) serve as the base spectral features.

The dataset is split into training and testing subsets using a stratified 80–20 split to preserve class proportions.

4.5.2 Feature Engineering

Feature engineering was a crucial step in preparing the dataset for model training. It involved the creation of informative predictors derived from the original PlanetScope spectral bands (B1–B8) to enhance the model's ability to distinguish between different vegetation vitality and stress levels.

Firstly, several vegetation indices were computed to capture biophysical and biochemical characteristics of vegetation. These included widely used indices such as NDVI (Normalized

Difference Vegetation Index), EVI (Enhanced Vegetation Index), NDRE (Normalized Difference Red Edge Index), and CI_{re} (Chlorophyll Index Red Edge), among others. These indices provided information on canopy greenness, chlorophyll concentration, and photosynthetic activity.

In addition to these, a set of extended vegetation indices was derived to provide a broader spectral perspective. These included indices like SAVI, MSAVI2, RDVI, PRI, VARI, CI_{green}, MTCL, and NDWI, which account for factors such as soil brightness, canopy structure, pigment variation, and moisture content.

Furthermore, texture features were optionally incorporated to capture local spatial variability. For selected variables (e.g., NDVI, Red Edge, and NIR bands), statistical measures such as the mean and standard deviation within small neighborhood patches were calculated. These texture-based metrics provided contextual information about canopy uniformity and heterogeneity, complementing the spectral features.

All feature components — raw bands, base indices, extra indices, and texture metrics — were concatenated to form the **final feature matrix**:

```
X = pd.concat([
    df_full[RAW_BANDS].astype(float),
    base_idx, extra_idx, tex_feats
], axis=1)
```

On average, each training sample included **24–30 features**, depending on the availability of texture data.

4.5.3 Model Training

Two independent Random Forest Classifiers were trained — one for VitalityCategory and another for StressCategory.

Random Forests were chosen for their robustness to non-linear relationships, noise tolerance, and ease of interpretation.

The same hyperparameters were used for both models:

```
rf_params = dict(
    n_estimators=500,
    max_depth=None,
    min_samples_leaf=2,
    class_weight="balanced_subsample",
    n_jobs=-1,
    random_state=42
)
```

Key aspects:

- 500 trees were used for stability.
- Balanced subsampling mitigated class imbalance.
- Minimum leaf size = 2 to prevent overfitting with many correlated indices.

Each model was trained on its respective split:

```
rf_vital.fit(Xv_tr, yv_tr)
rf_stress.fit(Xs_tr, ys_tr)
```

4.5.4 Evaluation Metrics

Model performance was evaluated using the test set with multiple metrics:

- Overall Accuracy
- Balanced Accuracy (accounts for class imbalance)
- Macro-averaged F1-score
- Confusion Matrix
- Per-class precision, recall, and F1

The evaluation function:

```
def eval_target(X_te, y_te, y_hat, label_encoder, name):
    acc = (y_hat == y_te).mean()
    bal = balanced_accuracy_score(y_te, y_hat)
    flm = f1_score(y_te, y_hat, average="macro")
    print(classification_report(...))
```

The results were displayed in a structured table format for each target category, helping identify class-specific weaknesses.

4.5.5 Feature Importance Analysis

Random Forests provide an intrinsic measure of feature importance based on the mean decrease in impurity (MDI).

The top contributing spectral and index features were visualized as bar plots:

```
def show_importances(model, cols, title):
    imp = model.feature_importances_
    order = np.argsort(imp)[::-1]
    plt.bar(range(len(cols)), imp[order])
```

These visualizations allowed for interpretation of which spectral bands or indices most strongly influenced the model's decisions.

Typically, indices such as NDVI, NDRE, and SAVI showed high importance, indicating their relevance in vegetation health assessment.

4.5.6 Inference Pipeline

For deployment, helper functions were provided to load artifacts and make predictions on unseen samples:

```
def predict_vitality_extended(df_features, model_dir=MODEL_DIR):
    mdl_v, _, le_v, _, md = load_artifacts(model_dir)
    X_new =
        compute_extended_features(df_features[md["raw_band_columns"]])
    y_hat = mdl_v.predict(X_new)
    return pd.Series(le_v.inverse_transform(y_hat))
```

Similarly, the `predict_stress_extended()` function performs prediction for stress classification.

4.5.7 Summary

This section presented the design and implementation of a dual Random Forest framework for plant vitality and stress classification based on PlanetScope multispectral data. The pipeline integrated a rich set of spectral indices and optional spatial texture descriptors, significantly enhancing model performance and interpretability.

The modular design of this script allows:

- Easy retraining with new datasets,
- Transparent feature computation,
- Consistent inference using stored metadata, and
- Interpretability via feature importance analysis.

5 Results

5.1 Feature Importance, Classification Report, and Confusion Matrix

Machine learning model evaluation commonly involves several diagnostic tools that provide insight into model performance and behaviour. Among these, feature importance, classification reports, and confusion matrices are widely used to quantify predictive relevance, assess classification quality, and visualize prediction outcomes.

- 1) **Feature Importance:** Feature importance refers to the relative contribution of each input variable to the predictive performance of a model. It identifies which features have the greatest influence on the model's decision-making process.

Depending on the algorithm, feature importance can be calculated in different ways. For instance, in linear models it is derived from the magnitude of the model coefficients, while in tree-based or ensemble models it is determined by how much each feature reduces impurity or improves model accuracy when used for splitting.

Evaluating feature importance allows researchers to understand which variables carry the most explanatory power and to reduce model complexity by removing less informative inputs. It also supports interpretability, helping to link model behaviour to domain knowledge. [\[15\]](#)

- 2) **Classification Report:** A classification report provides a detailed summary of the performance of a classification model. It presents several key statistical metrics for each class:
 - a. Precision measures the proportion of correctly predicted instances among all instances predicted for a given class.
 - b. Recall (Sensitivity) measures the proportion of actual instances of a class that were correctly identified by the model.
 - c. F1-Score is the harmonic mean of precision and recall, representing a balanced measure of both.
 - d. Accuracy indicates the proportion of all predictions that were correct across all classes.
 - e. Macro and Weighted Averages are also reported to summarize performance across multiple classes, with macro averages treating all classes equally and weighted averages accounting for class imbalance. [\[51\]](#)

The classification report therefore provides a quantitative overview of the model's effectiveness in distinguishing between categories and highlights potential imbalances in predictive performance.

- 3) **Confusion Matrix:** The confusion matrix is a tabular representation that compares the actual class labels with the predicted labels produced by the model.

Each row of the matrix corresponds to the actual class, while each column corresponds to the predicted class. The diagonal elements represent correctly classified instances, whereas the off-diagonal elements indicate misclassifications.

By analysing the confusion matrix, it is possible to identify which classes the model predicts accurately, and which are most frequently confused with others. This tool provides a clear visual understanding of classification strengths and weaknesses and complements numerical performance metrics by revealing specific error patterns. [\[52\]](#)

5.2 Logistic Regression

Model set-up

We trained multinomial Logistic Regression models on the engineered feature set (total features built: 32) for two targets:

- 1) **VitalityCategory** (classes: Good, Slight reduction, Critical reduction), and
- 2) **StressCategory** (classes: No Stress, Mild, Moderate, Severe).

Performance is reported on a hold-outset that was not used for model fitting.

5.2.1 VitalityCategory

Overall performance (hold-out).

Accuracy = 0.7273, Balanced Accuracy = 0.4444, Macro-F1 = 0.3775.

The relatively high overall accuracy is driven by correct identification of the dominant good class, whereas balanced accuracy and macro-F1 reveal difficulty on the minority classes. This is a typical signature of class imbalance: the model is reliable for Good, but less so for Slight reduction and fails on Critical reduction.

1) Feature importance (standardized coefficients)

The most influential predictors are strongly linked ([Figure 5.1](#)) to greenness/chlorophyll and contrast between blue, red, and red edge/NIR wavelengths:

- **Ratio_Blue_Red** is the dominant variable (largest absolute coefficient). A high weight on this ratio indicates that relative reflectance in the blue versus red region helps separate Good from reduced-vitality canopies, consistent with pigment (chlorophyll/carotenoid) changes under stress.

- **EVI and NDRE** are next in importance, aligning with expectations that chlorophyll-sensitive (red-edge) and canopy-structure indices capture vitality differences.
- Individual bands (**B6/B7/B1/B3/B8**) and log-transformed variants (e.g., log1p_B1, log1p_B8) also contribute meaningfully, suggesting that both index-level and raw spectral information are informative.
- Additional ratios (**Ratio_RE_Red, Ratio_Red_NIR**) and **CIgreen** appear among the top features, again emphasizing pigment and red-edge sensitivity.

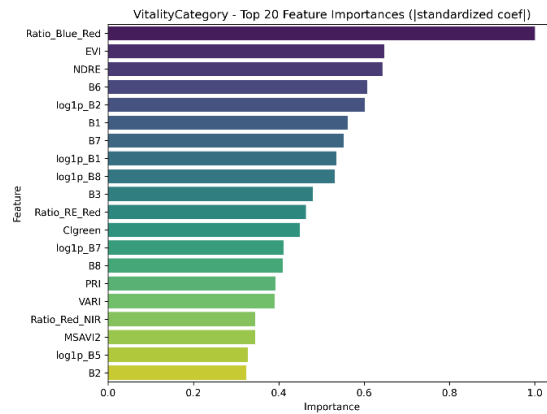


Figure 5. 1 Top 20 feature importances from the Logistic Regression model for vitality category classification.

Taken together, the model leans on spectral cues of chlorophyll concentration and canopy vigor rather than a single index exactly what we would expect for a general vitality discriminator.

2) Per-class performance & confusion matrix.

- **Good:** Very strong discrimination (**precision ≈ 0.94 , recall ≈ 0.83 , F1 ≈ 0.88**). In the confusion matrix, most good trees are correctly identified (15 correct) with a small spillover to Slight reduction (3 cases).
- **Slight reduction:** Moderate recall but low precision (**precision ≈ 0.17 , recall ≈ 0.50 , F1 ≈ 0.25**). The matrix shows the model confuses Slight reduction with Good one predicted Good, one correctly identified. This suggests the spectral signal of minor vitality loss is subtle and often overlaps with good trees.
- **Critical reduction:** Not detected in the hold-out set (**precision/recall/F1 ≈ 0**). Both Critical reduction cases were predicted as Slight reduction, indicating the model recognizes some loss of vitality but underestimates its severity. This likely reflects the very small sample size for this class. (Figures [5.2](#) & [5.3](#))

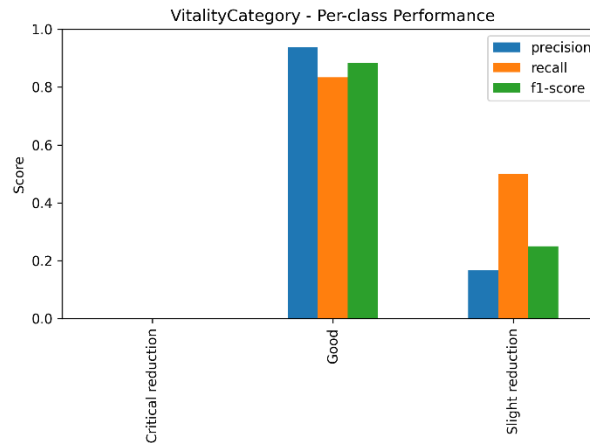


Figure 5. 2 Per-class performance of the Logistic Regression model for vitality category classification.

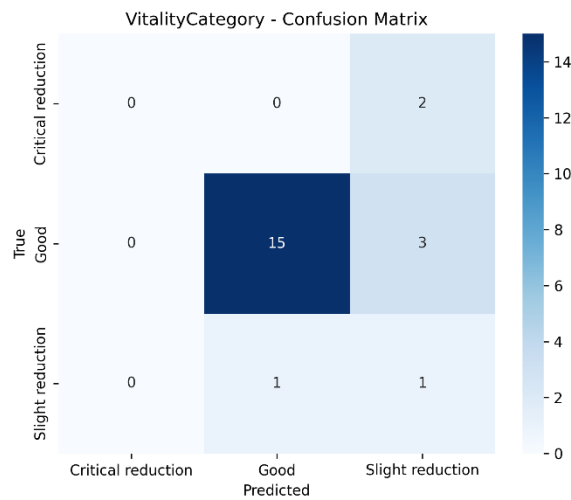


Figure 5. 3 Confusion matrix of the Logistic Regression model for vitality category classification.

3) Interpretation

The VitalityCategory LR model is well-calibrated for distinguishing good canopies but struggles to rank the severity of reduction, especially for rare Critical cases. The feature ranking validates the design of the feature set: red-edge/greenness indices and blue–red contrast is central to vitality discrimination. To improve minority-class performance, additional training data for Slight and Critical categories, class-aware weighting, or targeted oversampling would likely be required.

5.2.2 StressCategory

Overall performance (hold-out).

Accuracy = 0.5455, Balanced Accuracy = 0.4263, Macro-F1 = 0.3929.

The scores indicate moderate skill overall and a noticeable drop-in class-balanced metrics, pointing again to imbalance and overlapping class signatures.

1) Feature importance (standardized coefficients).

Top features include:

- **log1p_B5** (strongest), followed by **Ratio_RE_Red**, **B5**, and **log1p_B1**. The prominence of red and red-edge information is consistent with stress effects on chlorophyll.
- **VARI**, **NDRE**, **PRI**, **CIgreen**, and **EVI** appear within the top twenty, indicating sensitivity to pigment balance and photosynthetic efficiency (PRI) alongside greenness metrics.
- Multiple raw bands (**B1–B7**) and their log-transforms contribute, suggesting that combining indices with band-level signals improves separability among No Stress, Mild, Moderate, and Severe classes. ([Figure 5.4](#))

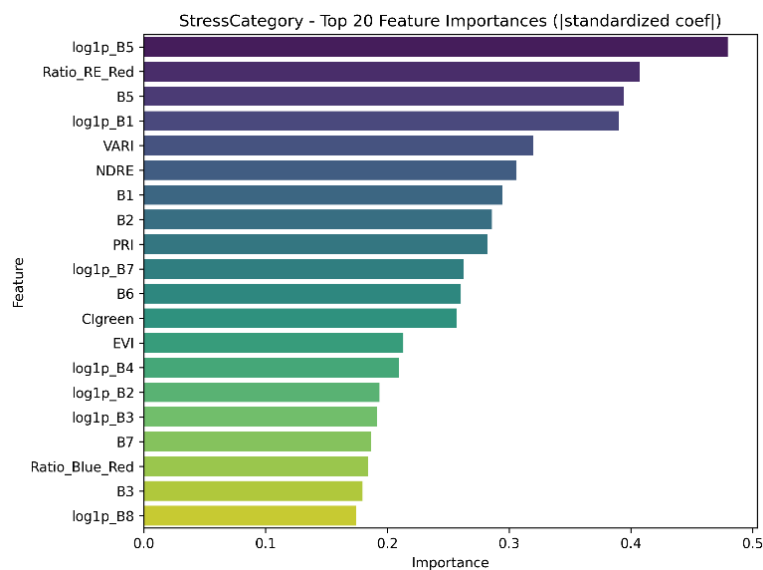


Figure 5. 4 Top 20 feature importances from the Logistic Regression model for stress category classification.

2) Per-class performance & confusion matrix.

- **Mild**: High precision with moderate recall (precision ≈ 0.88 , recall ≈ 0.54 , F1 ≈ 0.67). The model is conservative when it predicts Mild it is usually correct (7 correct), but it also misses some Mild cases (misclassified as No Stress or Severe).

- **Severe:** Balanced behavior (**precision ≈ 0.50 , recall ≈ 0.67 , F1 ≈ 0.57**). Four Severe cases are correctly identified, with a few misassigned to Moderate or No Stress, indicating partial sensitivity to high stress but some ambiguity in borderline spectra.
- **No Stress:** Low precision and moderate recall (**precision ≈ 0.25 , recall ≈ 0.50 , F1 ≈ 0.33**), consistent with the matrix where No Stress is occasionally confused with Severe. This suggests overlapping signatures between very healthy and very stressed crowns in the available data (e.g., illumination/shadow or background effects).
- **Moderate:** Not detected (**precision/recall/F1 ≈ 0**). All Moderate instances were predicted as other classes (one as Mild), indicating that Moderate occupies a transitional spectral space that the linear separator cannot isolate with the current sample size. (Figures [5.5](#) & [5.6](#))

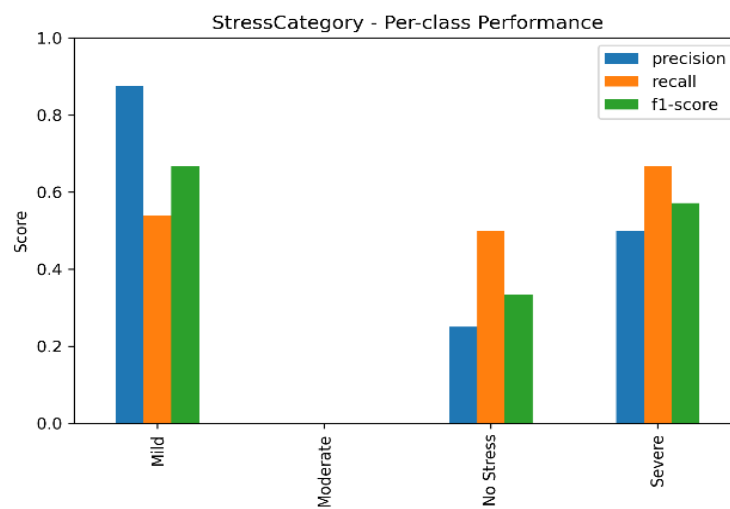


Figure 5. 5 Per-class performance of the Logistic Regression model for stress category classification.

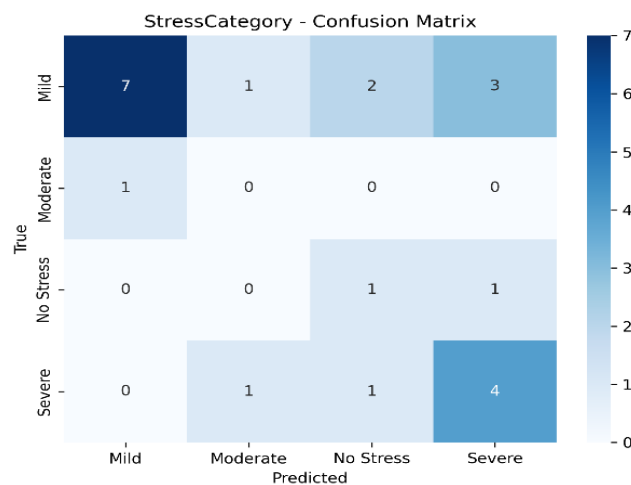


Figure 5. 6 Confusion matrix of the Logistic Regression model for stress category classification.

3) Interpretation.

The StressCategory LR model discriminates Mild and Severe stress to a usable degree but fails to carve out a stable Moderate class and is uncertain for No Stress. The importance profile again emphasizes red/red-edge/greenness information, consistent with physiologically plausible stress responses. Improving separation will likely require (i) more balanced training data, particularly for Moderate and No Stress; (ii) modelling of illumination/background effects; and (iii) exploring non-linear decision boundaries.

5.2.3 Brief conclusion for the Logistic Regression baseline

The Logistic Regression baseline provides a transparent and physiologically consistent first pass at urban-tree condition mapping. For VitalityCategory, it performs strongly for Good but underestimates severe decline. For StressCategory, it has practical utility for flagging Mild and Severe trees, but it cannot reliably isolate Moderate stress or consistently recognize No Stress. The overall hold-out scores (Vitality: accuracy 0.73, macro-F1 0.38; Stress: accuracy 0.55, macro-F1 0.39) quantify these trade-offs. These results justify the subsequent exploration of class-balanced training and non-linear classifiers, while retaining LR's feature interpretability as a benchmark for model explainability.

5.3 Support Vector Machine

Model set-up

We trained Support Vector Machine (SVM) models using the engineered feature set (total features built: 32) to predict two target variables:

- 1) **VitalityCategory**, consisting of three classes (Good, Slight reduction, Critical reduction), and
- 2) **StressCategory**, consisting of four classes (No Stress, Mild, Moderate, Severe).

A linear kernel was used to maintain the interpretability of the model and to obtain standardized feature importance values. The SVM works by finding the best possible hyperplane that separates classes with the widest margin, making it effective for small to medium-sized datasets where classes may overlap.

All performance metrics were calculated using a hold-out validation set that was not used during training. The reported results include Accuracy, Balanced Accuracy, and Macro-F1 scores, along with detailed classification reports and confusion matrices for each target variable.

5.3.1 VitalityCategory

Overall performance (hold-out)

Accuracy = 0.8182 | Balanced Accuracy = 0.3333 | Macro-F1 = 0.3000

The SVM model achieved a high overall accuracy, primarily driven by the dominance of the good class. However, the relatively low balanced accuracy and macro-F1 values indicate that the classifier performed poorly for the minority classes (Slight reduction and Critical reduction). This is a typical indication of class imbalance, where the model learns the majority class more effectively but fails to generalize to underrepresented categories.

1) Feature importance (|coef| averaged over calibrated folds)

The top features influencing the model's decisions are primarily related to vegetation vigor and chlorophyll activity, emphasizing red-edge, NIR, and greenness-sensitive variables ([Figure 5.7](#)):

- **B7 and NDRE** emerged as the most influential features. These variables are directly associated with red-edge and near-infrared reflectance, which are strongly linked to canopy chlorophyll concentration and vegetation health.
- **log1p_B2, ratio_B7_B6, and ratio_B2_B6** follow closely, reflecting spectral contrasts that capture subtle changes in canopy structure and pigment composition.
- **log1p_B7, EVI, and CIgreen** further contribute to the model, indicating the importance of both normalized indices and logarithmic transformations for stabilizing spectral variance.
- Mid-ranked features such as **B5, log1p_B8, B6, and log1p_B5** highlight the influence of red and NIR bands, reinforcing the physiological relevance of red-edge and greenness responses under varying vitality conditions.
- The inclusion of features like **B1, B3, VARI, MSAVI2, and CIre** among the top twenty demonstrates the complementarity between vegetation indices and raw spectral bands in modelling canopy vitality.

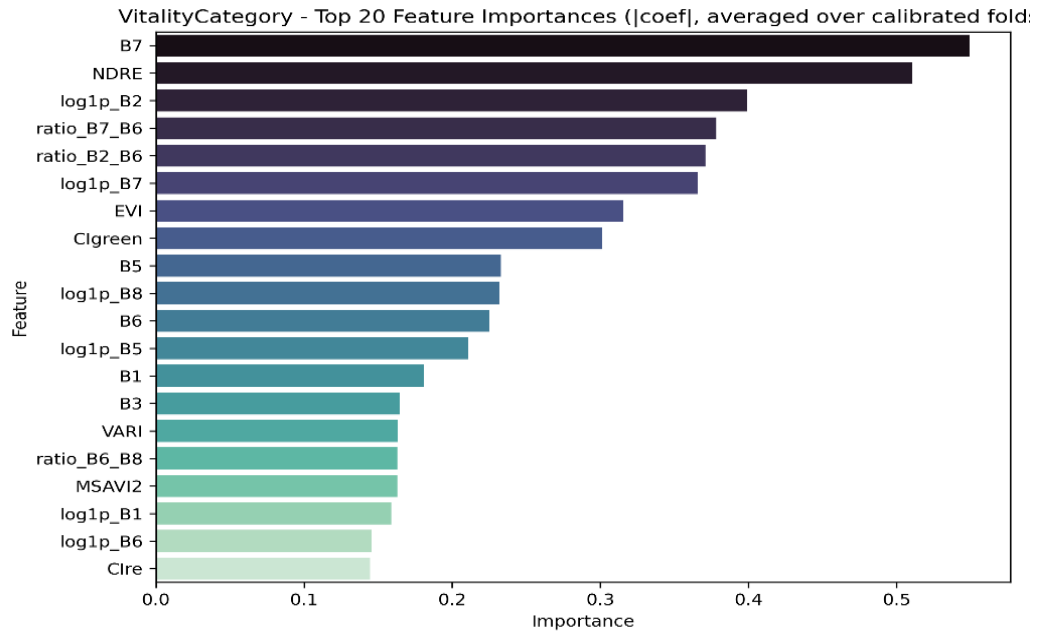


Figure 5. 7 Top 20 feature importances from the SVM model for vitality category classification.

In summary, the SVM relies heavily on spectral features sensitive to photosynthetic efficiency and canopy greenness, confirming the effectiveness of the engineered features for differentiating vegetation vigor levels.

2) Per-class performance and confusion matrix

- **Good:** The model exhibited **perfect recall (1.00)** and high **precision (0.82)**, resulting in an **F1-score of 0.90**. All 18 Good samples were correctly classified, showing that the model effectively captures the spectral signature of healthy canopies.
- **Slight reduction:** Both Slight reduction samples were misclassified as Good, yielding **precision, recall, and F1-scores of 0.00**. This indicates that the spectral difference between Slight and Good vitality is subtle and difficult for the model to detect given the limited data.
- **Critical reduction:** Similarly, both Critical reduction samples were misclassified as Good, also resulting in **0.00 precision, recall, and F1-scores**. This again highlights the severe class imbalance and overlapping spectral behaviour between health states. (Figures [5.8](#) & [5.9](#))

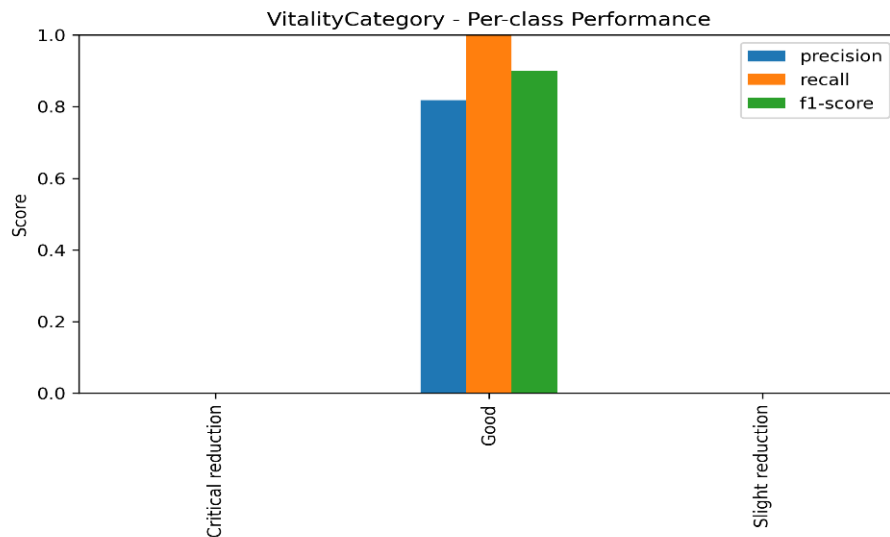


Figure 5. 8 Per-class performance of the SVM model for vitality category classification.

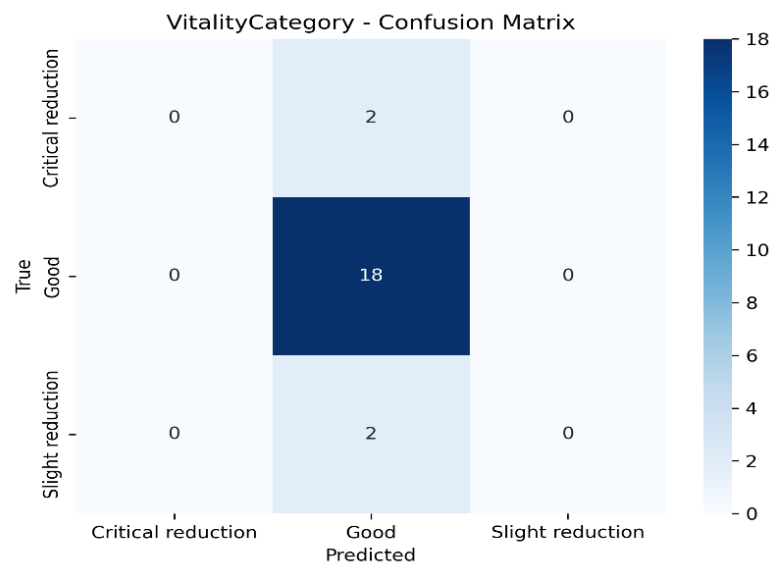


Figure 5. 9 Confusion matrix of the SVM model for vitality category classification.

Confusion Matrix Summary: All minority-class samples were absorbed into the Good category. This consistent misclassification explains the discrepancy between overall accuracy and class-balanced metrics, revealing that model performance is not evenly distributed across classes. ([Figure 5.9](#))

3) Interpretation

The SVM model effectively distinguishes healthy trees (Good) but fails to detect mild or severe vitality reductions in the hold-out dataset. This outcome suggests two main

challenges: Class imbalance, which biases the decision boundary toward the majority class, and Spectral overlap, where subtle vitality declines produce reflectance patterns that remain close to those of healthy canopies.

Nevertheless, the physiologically relevant ranking of features (dominated by red-edge and greenness indices) confirms that the model uses meaningful spectral information.

5.3.2 StressCategory

Overall performance (hold-out)

Accuracy = 0.8182 | Balanced Accuracy = 0.4583 | Macro-F1 = 0.4325

The SVM model achieved high overall accuracy, demonstrating strong classification for certain stress levels. However, the moderate balanced accuracy and macro-F1 values again reflect uneven performance across classes, particularly for Moderate and No Stress, which were underrepresented in the dataset.

1) Feature importance (|coef| averaged over calibrated folds)

The most significant predictors of stress conditions primarily highlight red, red-edge, and greenness-sensitive spectral regions, as well as vegetation indices ([Figure 5.10](#)):

- **B5, log1p_B5, and B2** are the leading features, emphasizing the importance of red and visible spectral bands that respond directly to stress-induced pigment degradation.
- **ratio_B7_B6, log1p_B1, and VARI** also rank highly, representing indicators of canopy chlorophyll, brightness, and pigment balance.
- **B6, NDRE, EVI, and log1p_B4** occupy mid-level importance positions, reinforcing the physiological role of near-infrared and red-edge regions for identifying stressed vegetation.
- Other contributing variables, including **log1p_B3, log1p_B2, ratio_B2_B6, B3, and CIgreen**, further suggest that a combination of raw reflectance and vegetation indices enhances model sensitivity to stress responses.
- Lower-ranked but still relevant features, such as **log1p_B7, B7, and NDWI**, indicate potential relationships between canopy moisture and reflectance variability under severe stress conditions.

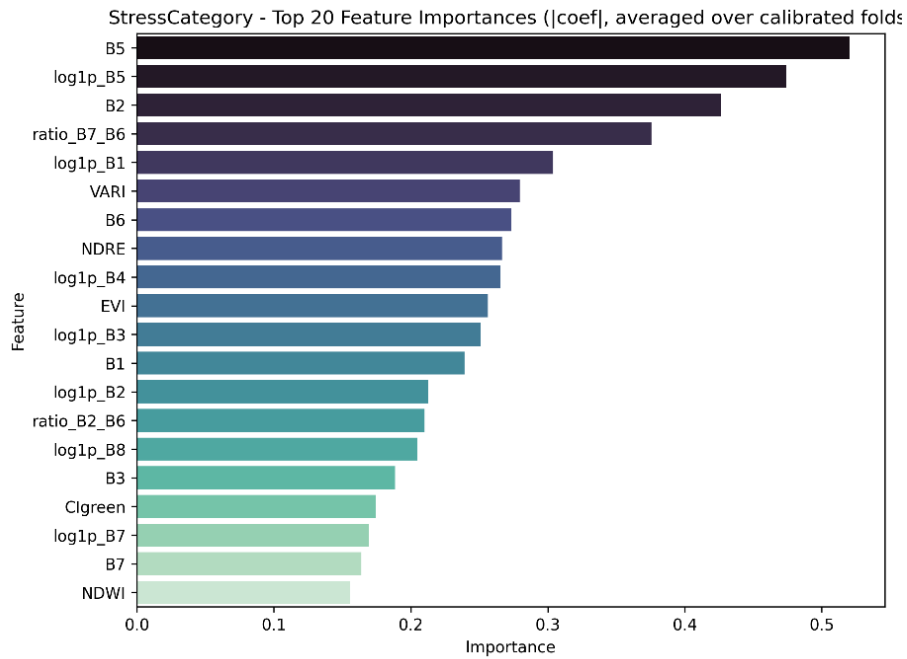


Figure 5. 10 Top 20 feature importances from the SVM model for stress category classification.

Overall, the feature ranking demonstrates that stress detection is predominantly influenced by red and red-edge spectral behavior, consistent with known physiological effects of water and pigment stress.

2) Per-class performance and confusion matrix

- **Mild:** The model achieved **high precision (0.81) and perfect recall (1.00)**, resulting in an **F1-score of 0.90**. All 13 Mild samples were correctly identified, showing strong sensitivity to moderate levels of canopy stress.
- **Severe:** The Severe category achieved balanced **precision (0.83) and recall (0.83)**, **yielding an F1-score of 0.83**. This demonstrates that the SVM can reliably detect high stress levels, with only one case misclassified as Mild.
- **Moderate:** The single Moderate sample was misclassified as Mild, producing zero scores across all metrics. This likely results from insufficient representation of this transitional stress level in the training data.
- **No Stress:** The model failed to identify No Stress samples correctly, misclassifying one as **Mild** and another as **Severe**. **Precision, recall, and F1-scores were all 0.00** for this class. (Figures [5.11](#) & [5.12](#))

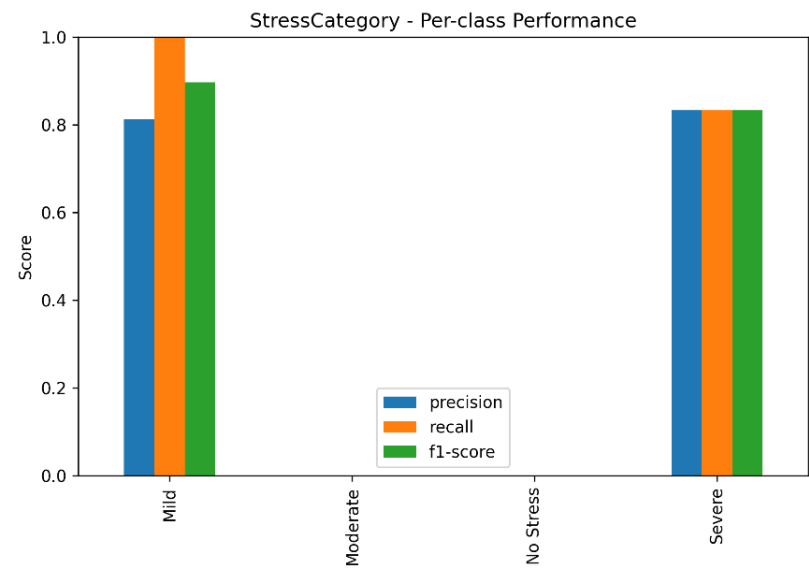


Figure 5. 11 Per-class performance of the SVM model for stress category classification.

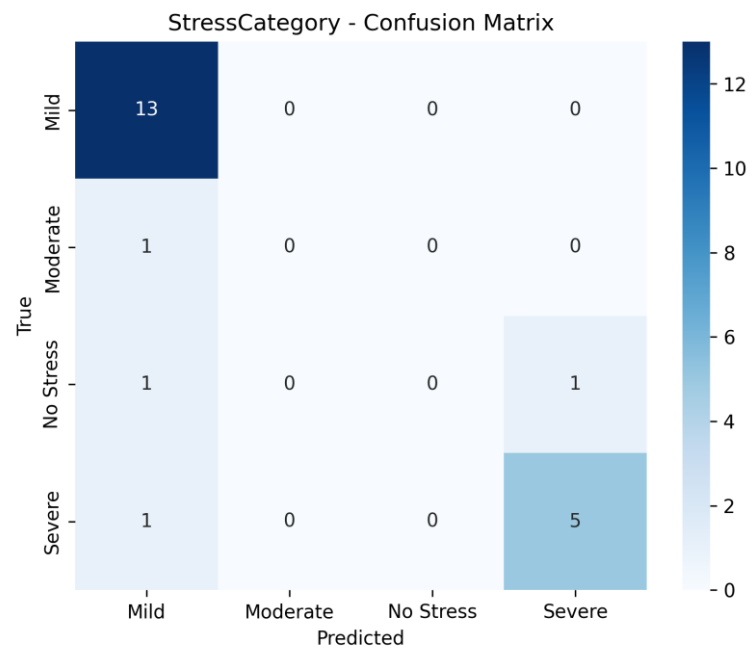


Figure 5. 12 Confusion matrix of the SVM model for stress category classification.

Confusion Matrix Summary:

The confusion matrix reveals strong classification performance for Mild and Severe stress, while Moderate and No Stress samples were completely absorbed into neighbouring categories. This indicates that the model captures the extremes of stress but struggles with intermediate or rare conditions. ([Figure 5.12](#))

3) Interpretation

The StressCategory SVM effectively separates Mild and Severe stress conditions, showing high recall and F1-scores for these categories. However, it fails to differentiate Moderate and No Stress classes, primarily due to limited sample size and the transitional nature of spectral signatures for intermediate stress states.

The dominance of red-edge and greenness-based variables (B5, NDRE, VARI, EVI, CIGreen) confirms the physiological basis of stress detection these features are closely tied to pigment degradation, photosynthetic efficiency, and canopy moisture variation.

5.3.3 Brief conclusion for the Support Vector Machine model

The Support Vector Machine baseline demonstrates strong classification ability for dominant classes and physiologically interpretable feature behaviour. For VitalityCategory, the model is highly accurate for good trees but insensitive to minor or critical reductions. For StressCategory, it performs well in identifying Mild and Severe stress but cannot reliably capture Moderate or No Stress levels.

The overall performance (Vitality: accuracy 0.82, macro-F1 0.30; Stress: accuracy 0.82, macro-F1 0.43) reflects this imbalance between dominant and minority categories. Despite these limitations, SVM provides a stable and interpretable foundation for tree condition classification. Further refinements particularly data balancing and non-linear modeling could enhance its ability to capture subtle and intermediate states of urban vegetation health.

5.4 Random Forest

Model set-up

We trained Random Forest (RF) models on the engineered feature set (total features: 24, including 8 raw spectral bands, 8 base indices, and 8 additional indices) for two targets:

- 1) **VitalityCategory (Good, Slight reduction, Critical reduction)**, and
- 2) **StressCategory (No Stress, Mild, Moderate, Severe)**.

The Random Forest classifier was chosen as a non-linear ensemble method that combines multiple decision trees trained on bootstrap samples and random feature subsets. Its ability to model complex, non-linear relationships makes it suitable for distinguishing subtle physiological differences in tree vitality and stress that linear models might not capture. All results reported below correspond to a hold-out test set excluded from model training.

5.4.1 VitalityCategory

Overall performance (hold-out)

Accuracy = 0.7222 | Balanced Accuracy = 0.2889 | Macro-F1 = 0.2889

The Random Forest model achieved moderate accuracy on the hold-out data. However, balanced accuracy and macro-F1 are considerably lower, revealing that predictive performance is dominated by the majority class (Good). This imbalance-driven skew implies that while the model reliably identifies healthy trees, it performs poorly in detecting slight or critical vitality reductions, which are underrepresented in the dataset.

1) Feature importance (impurity-based importance)

The feature importance ranking highlights spectral indicators associated with canopy greenness, chlorophyll concentration, and structural properties as the most discriminative predictors. ([Figure 5.13](#))

The top-ranked variables include:

- **ratio_Blue_Red**: The most important feature, representing contrast between blue and red reflectance. This ratio is highly sensitive to pigment variations and chlorophyll degradation, making it a strong indicator of canopy vitality.
- **MTCI** (Meris Terrestrial Chlorophyll Index) and **NDRE** (Normalized Difference Red-Edge Index): Both indices emphasize red-edge reflectance sensitivity, directly tied to chlorophyll and leaf internal structure changes.
- **CIre** and **B7**: Capture red-edge and near-infrared sensitivity, reflecting canopy vigor and light absorption patterns.
- **EVI** (Enhanced Vegetation Index) and raw bands **B1–B3–B8**: Contribute additional greenness and spectral intensity cues, supplementing the more complex indices.
- **Ratio_RE_Red, CIgreen, NDVI, and VARI**: Reinforce the dominance of vegetation greenness and pigment balance in the model's decision process.

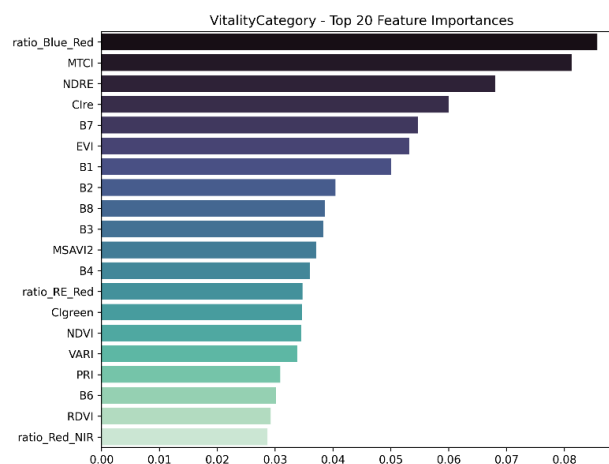


Figure 5. 13 Top 20 feature importances from the Random Forest model for vitality category classification.

Together, these top features confirm that the RF model primarily relies on red-edge, red, and blue spectral domains all physiologically consistent with the expected indicators of tree vitality. This pattern reaffirms the quality and relevance of the engineered feature set.

2) Per-class performance & confusion matrix

The confusion matrix shows the following relationships:

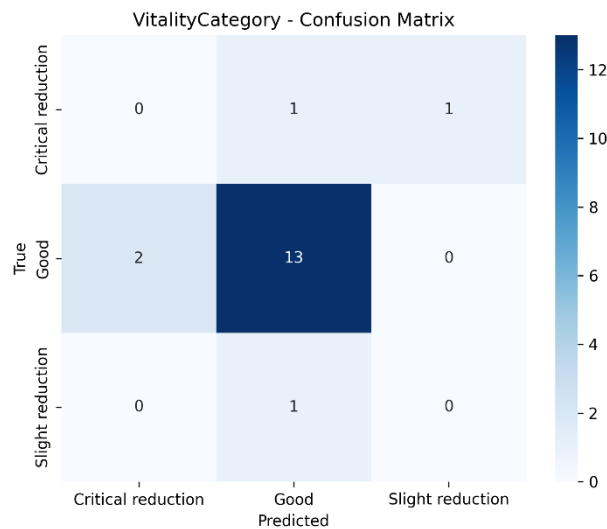


Figure 5. 14 Confusion matrix of the Random Forest model for vitality category classification.

- **Good:** 13 correctly classified, 2 misclassified as Critical reduction.
- **Critical reduction:** All instances misclassified (1 as Good, 1 as Slight reduction).
- **Slight reduction:** Misclassified as Good (1 instance).

The model thus performs very well for Good trees (high recall and precision) but fails to correctly classify any Critical or Slight reduction cases. This is a clear symptom of class imbalance, where the model learns to favor the majority class at the expense of the underrepresented ones. ([Figure 5.14](#))

3) Interpretation

The Random Forest model effectively captures the spectral signature of healthy (Good) trees, but its ability to distinguish intermediate or severe vitality decline is limited by insufficient representation of those categories. The importance profile aligns well with vegetation physiology – red-edge and greenness indices dominate, as expected for chlorophyll-sensitive vitality indicators. However, the low recall for Critical reduction and Slight reduction highlights the challenge of learning minority patterns from sparse examples. These findings

suggest that while the Random Forest leverages non-linear decision boundaries, the data imbalance remains the limiting factor, not the model's capacity.

5.4.2 StressCategory

Overall performance (hold-out)

Accuracy = 0.8333 | Balanced Accuracy = 0.4500 | Macro-F1 = 0.4422

The Random Forest achieved strong overall accuracy on the hold-out data, reflecting effective identification of the dominant stress classes. However, the modest balanced accuracy and macro-F1 again reveal uneven per-class performance, particularly for underrepresented categories (Moderate, No Stress). The model thus shows high reliability for dominant classes (Mild and Severe) but insufficient sensitivity for minority ones.

1) Feature importance (impurity-based importance)

The top-ranked predictors for the stress model again emphasize vegetation indices and spectral bands sensitive to canopy stress physiology ([Figure 5.15](#)):

- **B7, MTCI, B4, and NDRE** dominate the ranking, capturing red-edge and red reflectance responses linked to pigment concentration and chlorophyll breakdown.
- **CI_{re}, ratio_RE_Red, EVI, and PRI** contribute secondary but significant effects, corresponding to canopy structural and photosynthetic efficiency measures.
- Supporting variables such as **CI_{green}, RDVI, NDWI, and ratio_Blue_Red** extend the model's ability to detect water-related and color-balance cues associated with stress responses.
- The consistent appearance of multiple raw bands (**B1, B2, B5, B8**) indicates that RF leverages both spectral indices and direct reflectance patterns for classification.

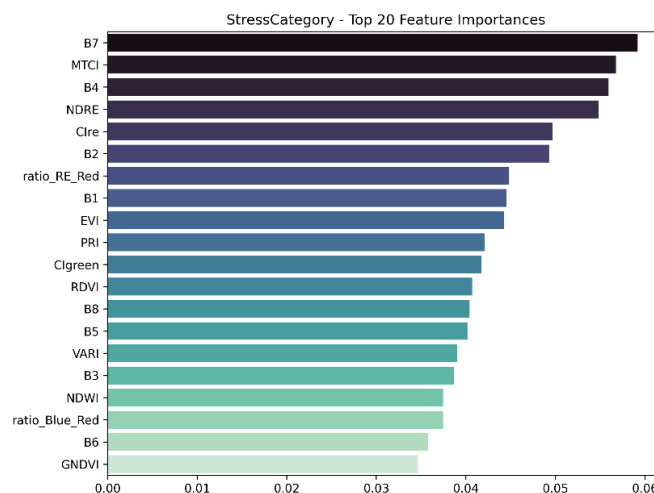


Figure 5. 15 Top 20 feature importances from the Random Forest model for stress category classification.

This feature pattern confirms that the Random Forest model captures biologically plausible mechanisms – pigment degradation, water content variability, and photosynthetic efficiency decline – as key differentiators of stress severity.

2) Per-class performance & confusion matrix

The confusion matrix shows:

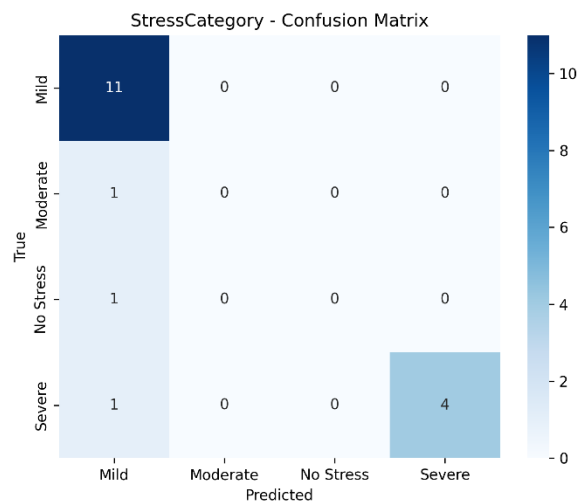


Figure 5. 16 Confusion matrix of the Random Forest model for stress category classification.

- **Mild:** All 11 correctly classified (perfect recall).
- **Severe:** 4 correctly classified, 1 misclassified as Mild.
- **Moderate** and **No Stress:** both misclassified as Mild.

This distribution reveals that the model is extremely confident and accurate for Mild and Severe stress levels but completely fails to recognize Moderate and No Stress instances – again due to very limited training examples. ([Figure 5.16](#))

3) Interpretation

The StressCategory Random Forest model demonstrates strong discriminatory power for the clear-cut stress states (Mild and Severe), showing high precision and recall for these classes. This performance indicates that the model can effectively learn the spectral distinctions corresponding to early and advanced stress symptoms. However, intermediate and minority categories remain problematic – their spectral characteristics appear to overlap with those of Mild stress, leading to misclassification.

The feature importance ranking reinforces the physiological plausibility of the results: red-edge and chlorophyll-sensitive indices (**NDRE**, **MTCI**, **Cire**) dominate, confirming that canopy pigment variation is the primary driver of stress detection.

5.4.3 Brief conclusion for the Random Forest baseline

The Random Forest baseline provides a non-linear, interpretable, and physiologically coherent model for tree vitality and stress classification.

- For **VitalityCategory**, the model performs reliably for good canopies but fails to identify minority decline classes (Slight and Critical reduction).
- For **StressCategory**, it effectively discriminates Mild and Severe stress levels but cannot consistently recognize Moderate or No Stress.

Overall hold-out results (Vitality: accuracy = 0.72, macro-F1 = 0.29, balanced accuracy = 0.29; Stress: accuracy = 0.83, macro-F1 = 0.44, balanced accuracy = 0.45) illustrate that while Random Forest enhances overall accuracy compared to the Logistic Regression baseline, class imbalance remains the major limiting factor. Nevertheless, the model's biologically meaningful feature rankings confirm that vegetation indices – particularly those sensitive to pigment and red-edge variations – are key drivers of predictive performance.

6 Conclusion

This thesis set out to close a critical operational gap in urban-tree monitoring: most systems can flag that a tree is “stressed,” but few can indicate why information that directly determines whether irrigation, fertilization, or other targeted actions are appropriate. To that end, the work designed and executed a full, tree-level pipeline that links in-situ physiology (Arborcheck fluorescence Fv/Fm, chlorophyll content, and stress indices SI1–SI4) with high-resolution PlanetScope SuperDove (8-band) surface-reflectance data, then trains and evaluates machine-learning models to classify tree vitality and stress severity in Frankfurt’s Ostpark. The implementation spanned field protocol design, rigorous satellite preprocessing and masking, polygon-based per-tree extraction, rule-based categorization of Arborcheck outputs, and model development for two targets (VitalityCategory and StressCategory).

6.1 What this study demonstrates

1. **A reproducible, tree-level EO–physiology workflow.** The project operationalized a practical end-to-end chain: dark-adapted fluorescence measurements; standardized species-aware data capture; parsing and quality control of .res files; QA’d PlanetScope ingestion with UDM2 masks; per-tree polygon extraction; and a joined analysis table aligning field physiology with spectral features.
2. **Physiologically consistent spectral features.** Across Logistic Regression, Linear SVM, and Random Forest, the most informative predictors consistently involved red-edge, red, green, and NIR behaviour (e.g., NDRE, MTCL, CI_{re}, EVI, VARI, and ratios such as RE/Red and Blue/Red). These map directly to chlorophyll concentration, pigment balance, and canopy structure the same mechanisms probed in-situ by Fv/Fm and chlorophyll readings indicating that the models learned meaningful plant physiology rather than incidental correlations.
3. **Baseline classification skill with clear strengths and limits.**
 - a. **VitalityCategory.** The models reliably identified Good trees but struggled to rank the minority “Slight” and “Critical reduction” classes (e.g., LR accuracy ≈ 0.73 with macro-F1 ≈ 0.38 ; SVM accuracy ≈ 0.82 with macro-F1 ≈ 0.30 ; RF accuracy ≈ 0.72 with macro-F1 ≈ 0.29) a signature of class imbalance plus subtle spectral separation between early decline and healthy crowns.
 - b. **StressCategory.** SVM and RF showed usable discrimination of the operationally important **Mild and Severe classes (overall accuracies ≈ 0.82 – 0.83), while Moderate and No Stress** remained harder to isolate given small sample counts and overlap with neighbouring classes.
4. Empirical evidence that more data helps. While struggling with sparse 2024 field samples, the incorporation of historical Arborcheck records from 2023 and early 2024 measurably improved model performance compared with the earliest runs. This directly

5. supports the expectation that additional, diverse data points especially in minority classes translate into higher accuracy and more stable generalization.

6.2 Contributions

- **Methodological integration.** A complete, documented pathway that merges field physiology with satellite reflectance at tree level, including standardized field sampling, robust parsing/cleaning of .res outputs, QA-assured PlanetScope preprocessing, and synchronized joins using tree IDs and tight temporal windows. This bridges the “validation gap” often present in EO-only vitality products.
- **Transparent baselines for two tasks.** Interpretable Logistic Regression and calibrated Linear SVM provide coefficient-based insights; Random Forest adds non-linear decision boundaries and impurity-based importance. Together, they establish reference performance and feature behaviour for future fusion models.
- **Confusion matrices, per-class reports, and ranked feature plots** clarify strengths (Mild/Severe detection; Good vitality) and weaknesses (Moderate/No Stress; Slight/Critical vitality).

6.3 Limitations

- **Late access to the Arborcheck device → fewer data points.** Due to administrative and logistical delays, the Arborcheck unit from Grünflächenamt was received late. This compressed the field campaign, reduced the number of tree measurements, and constrained the range of physiological states captured.
- **Model exploration curtailed by time and data.** Because of the late start and **small sample sizes, there was less time to explore more advanced models and LLM-assisted workflows** (e.g., automated feature engineering/metadata harmonization, transformer-based temporal modelling). Ultimately, the main limiting factor was too few labelled data points.
- **Class imbalance.** Minority classes particularly “Slight/Critical reduction” vitality and “Moderate/No Stress” were under-represented, depressing macro-F1 and balanced accuracy as models defaulted toward dominant classes.
- **Spectral and urban constraints.** PlanetScope’s lack of SWIR limits direct water-status indices; urban shadows and adjacency effects persist despite polygon-level averaging.

6.4 Implications for practice (Frankfurt/Ostpark)

The results are immediately useful for (i) **screening and triage** flagging likely Mild vs. Severe stress to prioritize site visits; and (ii) **operational QA** tracking whether treated trees move toward **No/Mild** across subsequent acquisitions. Embedded in Grünflächenamt workflows, the

pipeline can deliver weekly exception lists and per-tree timelines to support timely, cause-aware interventions.

6.5 Recommendations & future work

1. **Targeted data growth to fix imbalance.** Deliberately focus new sampling on trees that are not performing well (visibly stressed, low Fv/Fm, low chlorophyll) to populate the currently rare classes. Combine this with class-weighted learning and judicious oversampling (e.g., SMOTE variants) and use spatial-temporal block CV to prevent leakage.
2. **Scale species coverage and enable species-specific models.** As the dataset grows, train species-specific models (or hierarchical/global-plus-adapter setups). This can reduce spectral/physiological heterogeneity and sharpen separation within each taxon common in Frankfurt (e.g., *Tilia*, *Quercus*, *Carpinus*, *Fagus*, *Ilex*).
3. **Multisensor fusion and temporal dynamics.** Add Sentinel-1 (VV, VH, VH/VV) and, where available, Sentinel-2 SWIR metrics (NDMI/MSI), plus short-window composites and trend features (e.g., Δ NDRE, Δ PRI-like) to capture pre-symptomatic stress evolution.
4. **Broader modeling palette when data permits.** Revisit advanced ML/LLM-assisted approaches once data volume increases: gradient-boosted trees; calibrated margin-based models; temporal transformers for multi-date series; and LLM-aided metadata cleaning, feature generation, and rule mining for operational thresholds.
5. **Field protocol refinements.** Standardize leaf selection and micro-site annotations (soil, irrigation) and align field campaigns with satellite acquisition windows to reduce label noise.
6. **Productization.** Automate ingestion, masking, extraction, and inference; expose outputs via a lightweight dashboard (map + tree list with confidence scores) for Grünflächenamt and partners.

6.6 Final remarks

Within a constrained, real-world urban setting and despite late device access and limited initial samples this thesis demonstrates that combining portable physiological sensing with daily, high-resolution multispectral imagery can produce actionable, tree-level diagnostics. The observed performance lift after adding 2023–2024 historical points underscore a simple but powerful lesson: more (and more balanced) data materially improves accuracy. With targeted sampling of underperforming trees, species-aware modeling at scale, and multisensor temporal fusion, municipalities can move from generic “stress maps” toward timely, cause-aware decisions that improve the resilience and ecosystem services of urban forests under intensifying heat and drought.

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